

Smarter Machine Building **Starts Here**



Smarter Machine Building Starts Here

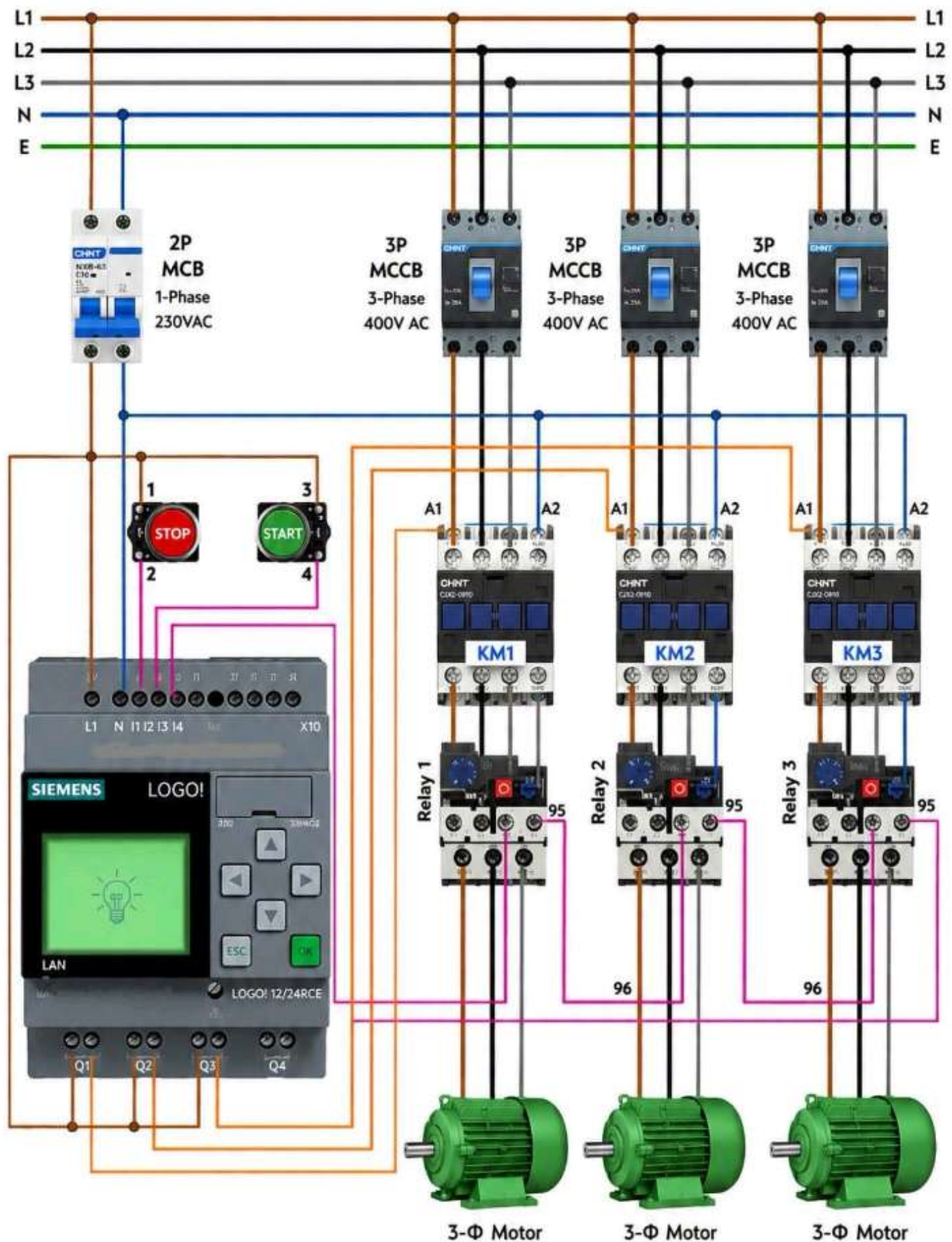
ACCESS GUIDE



ing

u face as a
ts and develop
e long term.
your machines.
making a

Sequential Operation of 3- Φ Motors Using LOGO! V8 PLC



Stop Configuring MikroTik One by One.

Standardize. Automate. Scale.

OptiWize helps MSPs and system integrators manage all your MikroTik devices from one centralized platform — templates, backups, monitoring and more.



One Template.
Hundreds of Devices.



Automated Backups.
Always Protected.



Monitor & Troubleshoot.
Remotely.



Smart Alerts.
Instant Visibility.



Save
Time



Reduce
Errors



Scale With
Confidence



Secure &
Reliable



Deliver Better
for Your Clients

[SEE HOW IT WORKS →](#)

Centralize your MikroTik operations
and focus on growing your business.

BASIC KNOWLEDGE FOR CIVIL SITE ENGINEERS

Civil Engineers 07

- **Weight of steel = $D^2 / 162$ (kg / m)**
- **Weight of steel = $D^2 / 533$ (kg / ft)**
- **Size of concrete testing cube = 150 x 150 x 150 mm**
- **Concrete cube is filled in = 3 layers**
- **Slump cone is filled in = 4 layers**
- **Minimum % of steel in column = 0.8% of gross area**
- **Maximum % of steel in column = 6% of gross area**
- **Minimum % of steel used in beam = 1% of gross area**
- **Maximum % of steel used in beam = 2% of gross area**
- **Minimum % of steel used in slab = 0.7% of gross area**
- **Maximum % of steel used in slab = 1% of gross area**
- **Minimum thickness of shear wall = 150mm**
- **Maximum thickness of shear wall = 400mm**
- **Slope of staircase = 25 to 40 degree**
- **Nominal Cover for :- Footing = 50mm**
Column = 40mm
Slab = 20mm



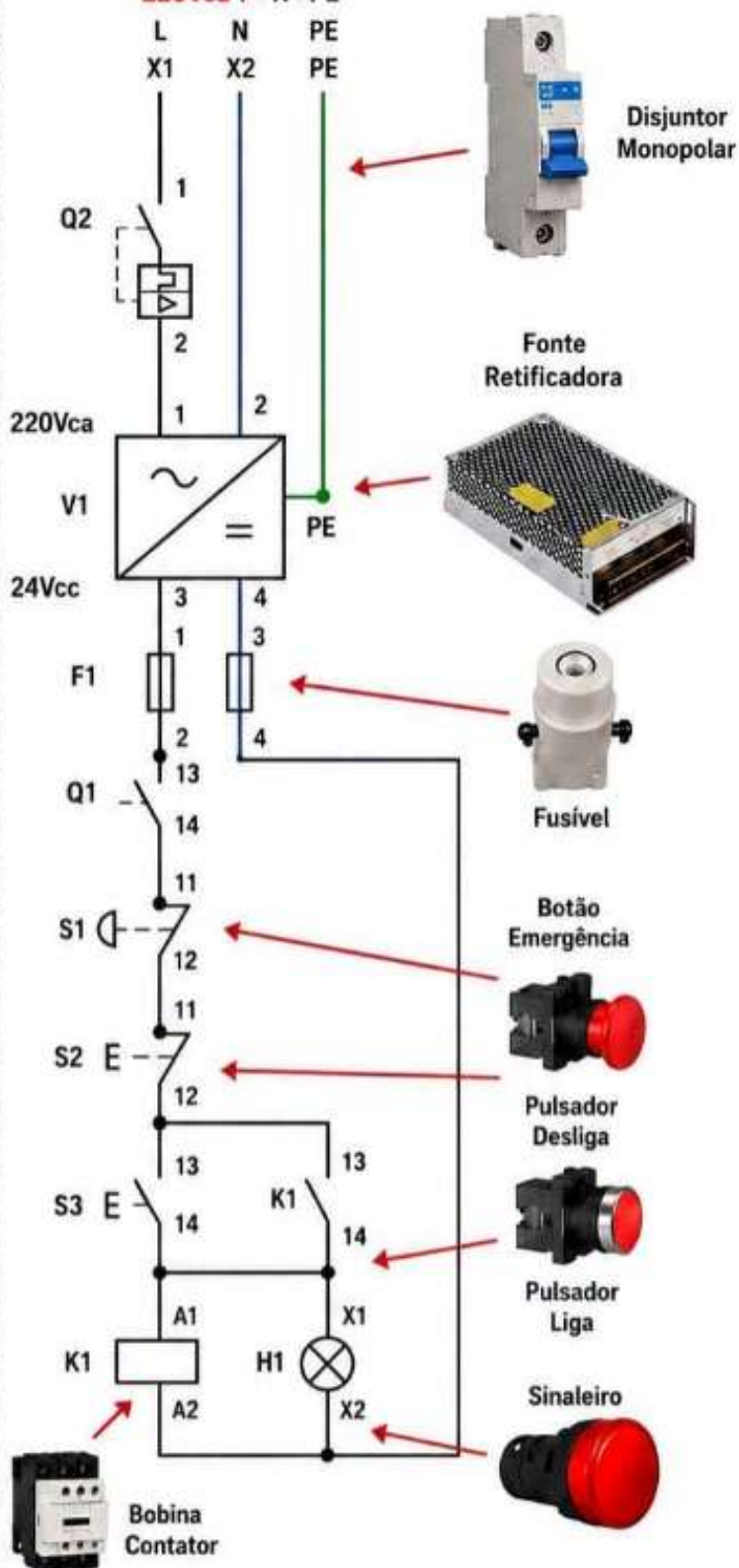
CIRCUITO DE FORÇA

440Vca 3F



CIRCUITO DE COMANDO

220Vca F+N+PE



LEGENDA DE TERMINAIS (ATS)

1	2	3	4	— FONTE A (NORMAL)
L1	L2	L3	N	
5	6	7	8	— CARGA
L1	L2	L3	N	
9	10	11	12	— FONTE B (EMERGÊNCIA)
L1	L2	L3	N	

CÓDIGO DE CORES

—	L1 (VERMELHO)
—	L2 (AMARELO)
—	L3 (AZUL)
—	N (NEUTRO)
—	PE (TERRA)

FUNÇÕES DOS COMPONENTES

Q1	Disjuntor Motor	S2	Pulsador Desliga (NF)
K1	Contator de Força	S3	Pulsador Liga (NA)
Q2	Disjuntor Monopolar	K1	Bobina Contator
F1	Fusível	H1	Sinalizador
S1	Botão Emergência (NF)	V1	Fonte Retificadora



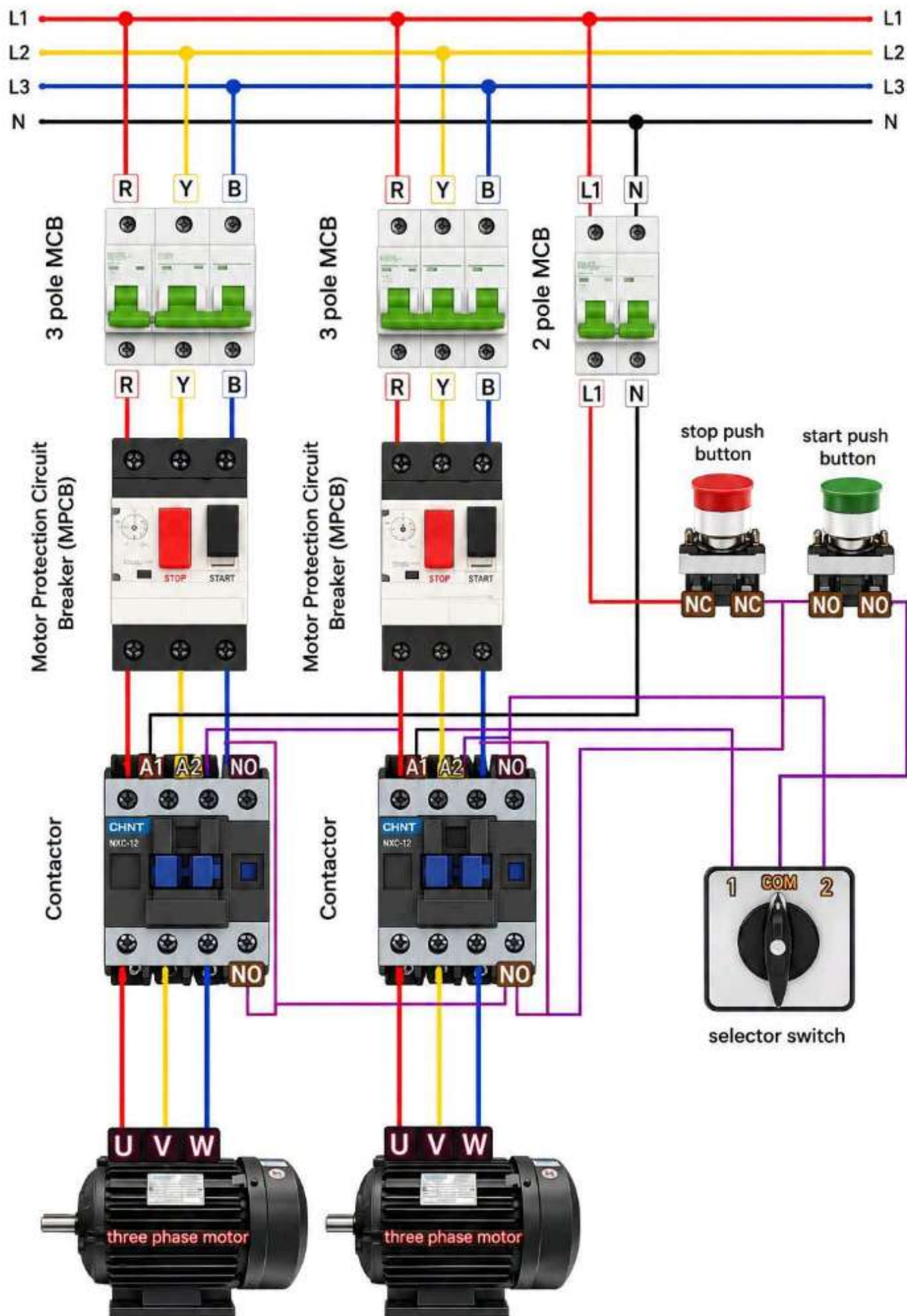
ATENÇÃO:

Certifique-se de que todos os equipamentos estejam desenergizados antes da instalação.

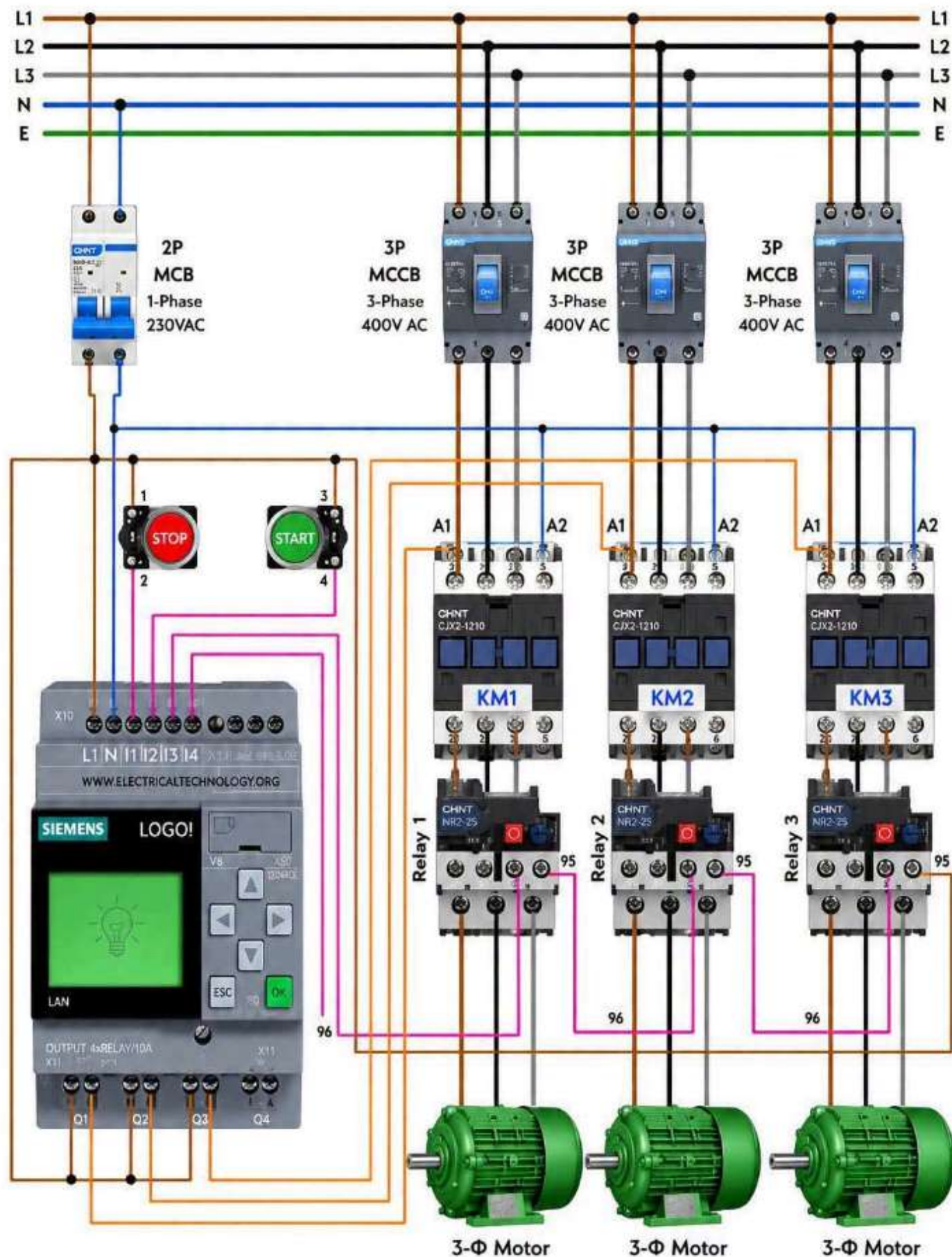


ATERRAMENTO:

Conecte o PE (terra) em todos os equipamentos metálicos.



Sequential Operation of 3- Φ Motors Using LOGO! V8 PLC





ALL ESSENTIAL ELECTRICAL Electrical FULL FORMS Knowledge

BASIC ELECTRICAL QUANTITIES

- V = Volt
- I = Ampere
- R = Ohm
- W = Watt
- VA = Volt-Ampere
- VAR = Volt-Ampere Reactive
- PF = Power Factor
- Hz = Hertz
- F = Farad
- H = Henry
- S = Siemens
- C = Coulomb
- J = Joule
- T = Tesla

ELECTRICAL CIRCUITS & AC

- AC = Alternating Current
- DC = Direct Current
- RMS = Root Mean Square
- P = Power
- Q = Reactive Power
- S = Apparent Power
- PF = Power Factor
- Z = Impedance
- X_L = Inductive Reactance
- X_C = Capacitive Reactance
- f = Frequency
- ω = Angular Frequency
- ϕ = Phase Angle

ELECTRICAL ELEMENTS

- R = Resistor
- L = Inductor
- C = Capacitor
- D = Diode
- LED = Light Emitting Diode
- TR = Transformer
- SW = Switch
- CB = Circuit Breaker
- FUSE = Fuse
- NR = Non-Return (Diode)
- SCR = Silicon Controlled Rectifier
- UJT = Unijunction Transistor

UNITS & MEASUREMENTS

- V = Volt
- A = Ampere
- Ω = Ohm
- W = Watt
- kW = Kilowatt
- MW = Megawatt
- kWh = Kilowatt-hour
- MWh = Megawatt-hour
- VAR = Volt-Ampere Reactive
- VA = Volt-Ampere
- Hz = Hertz
- F = Farad
- H = Henry
- S = Siemens
- T = Tesla

ELECTRICAL MACHINES

- DCM = DC Machine
- ACM = AC Machine
- IM = Induction Motor
- SM = Synchronous Motor
- GM = Generator (General)
- SRM = Switched Reluctance Motor
- BLDC = Brushless DC Motor
- PMDC = Permanent Magnet DC Motor
- PWM = Pulse Width Modulation
- VFD = Variable Frequency Drive

POWER SYSTEMS

- GEN = Generator
- TRF = Transformer
- TX = Transmission
- DISTR = Distribution
- HT = High Tension
- LT = Low Tension
- EHV = Extra High Voltage
- HV = High Voltage
- MV = Medium Voltage
- LV = Low Voltage
- PT = Potential Transformer
- CT = Current Transformer

PROTECTION EQUIPMENT

- CB = Circuit Breaker
- MCB = Miniature Circuit Breaker
- MCCB = Molded Case Circuit Breaker
- RCCB = Residual Current Circuit Breaker
- ELCB = Earth Leakage Circuit Breaker
- FUSE = Fuse
- OC = Overcurrent
- SC = Short Circuit
- CT = Current Transformer
- PT = Potential Transformer

ELECTRICAL INSTALLATION

- DB = Distribution Board
- SMDB = Sub Main Distribution Board
- MDB = Main Distribution Board
- SP = Single Pole
- DP = Double Pole
- TP = Three Pole
- SPDT = Single Pole Double Throw
- DPDT = Double Pole Double Throw
- G = Ground / Earth
- N = Neutral
- P = Phase

WIRES & CABLES

- CU = Copper
- AL = Aluminum
- PVC = Polyvinyl Chloride
- XLPE = Cross Linked Polyethylene
- SWA = Steel Wire Armoured
- FRLS = Flame Retardant Low Smoke
- NYA = Non-sheathed Cable
- NYCY = PVC Insulated, PVC Sheathed Cable
- ABC = Aerial Bunched Cable

ELECTRONICS

- IC = Integrated Circuit
- PCB = Printed Circuit Board
- MIC = Microphone
- LED = Light Emitting Diode
- LCD = Liquid Crystal Display
- MOSFET = Metal Oxide Semiconductor Field Effect Transistor
- BJT = Bipolar Junction Transistor
- OP-AMP = Operational Amplifier

GENERAL

- AM = Amplitude Modulation
- FM = Frequency Modulation
- PM = Phase Modulation
- R&D = Research and Development
- PLC = Programmable Logic Controller
- UPS = Uninterruptible Power Supply
- SCADA = Supervisory Control And Data Acquisition

IMPORTANT SIGNS

- $\sim$ = AC (Alternating Current)
- $---$ = DC (Direct Current)
- $\oplus$ = Earth / Ground
- + = Positive
- - = Negative

QUICK REFERENCE CONSTANTS

- | | | | |
|---|-------------------------------|-------------------------------|-------------------------------|
| • Speed of Light (c) | = 3.00×10^8 m/s | • Elementary Charge (e) | = 1.602×10^{-19} C |
| • Permeability of Free Space (μ_0) | = $4\pi \times 10^{-7}$ H/m | • Planck Constant (h) | = 6.626×10^{-34} J·s |
| • Permittivity of Free Space (ϵ_0) | = 8.854×10^{-12} F/m | • Avogadro Constant (N_A) | = 6.022×10^{23} /mol |
| • Boltzmann Constant (k) | = 1.381×10^{-23} J/K | | |

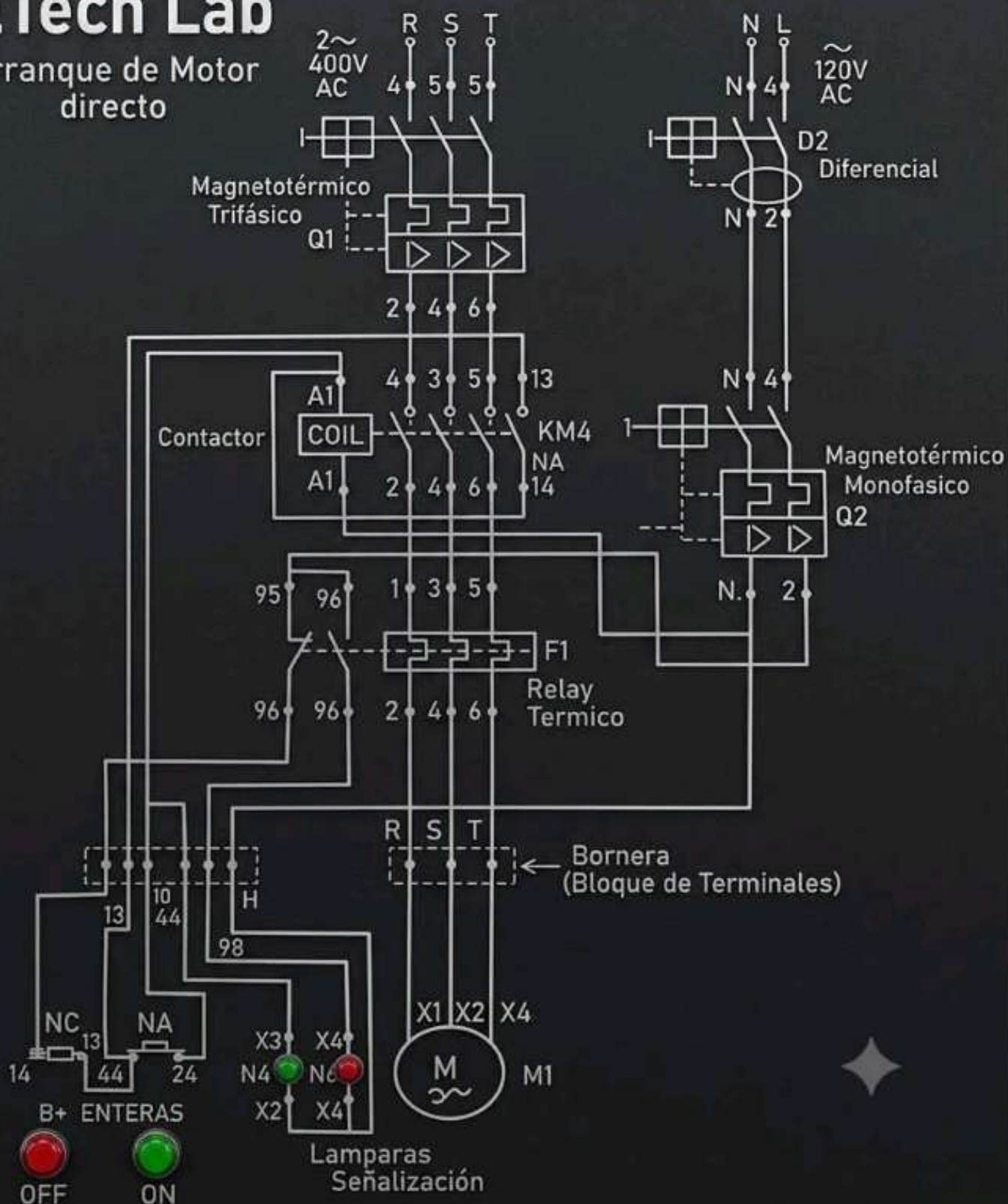
MEMORY TIPS

- | | | |
|------------------------|----------------------------------|--------------------|
| ★ V R → Ohm's Law | ★ S = V × I (AC Apparent Power) | ★ 1 kW = 1000 W |
| ★ P = V × I (DC Power) | ★ PF = cos ϕ (Power Factor) | ★ 1 MWh = 1000 kWh |



ETech Lab

Arranque de Motor directo



HOW TO READ
CONSTRUCTION DRAWINGS

A SIMPLE GUIDE FOR CIVIL ENGINEERS & STUDENTS



1 TITLE BLOCK

Always check the title block first. It gives important information about the project.

- ✓ Project Name
- ✓ Drawing Title
- ✓ Drawing No.
- ✓ Scale
- ✓ Date
- ✓ Designed / Drawn by

PROJECT	RESIDENTIAL BUILDING
DRAWING TITLE	GROUND FLOOR PLAN
DRAWING NO.	A-01
SCALE	1:100
DATE	15-06-2025
DESIGNED BY	ENGR. RAHUL
DRAWN BY	ENGR. KAPIL
CHECKED BY	ENGR. SAHIL

2 SCALE

Scale shows the ratio between drawing and actual size.

COMMON SCALES

1 : 50	1 cm = 0.5 m
1 : 100	1 cm = 1 m
1 : 150	1 cm = 1.5 m
1 : 200	1 cm = 2 m



Note: Always use scale to find actual dimensions.

3 SYMBOLS

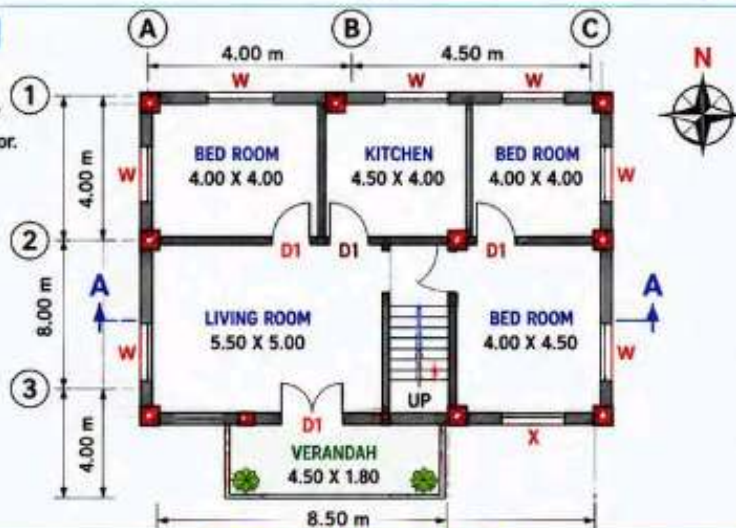
Learn basic symbols used in drawings.

	Column		Staircase
	Brick Wall		Center Line
	Door		Section Mark
	Window		Level Mark
	Ventilator		

4 PLAN (FLOOR PLAN)

Plan shows the layout of the building as if we cut it horizontally at about 1.2 m height from the floor.

- ✓ Rooms
- ✓ Walls
- ✓ Doors
- ✓ Windows
- ✓ Columns
- ✓ Dimensions
- ✓ etc.



GROUND FLOOR PLAN

5 DIMENSIONS

Dimensions show the size of different elements.

- ✓ Read in meters (m)
- ✓ Always follow overall dimensions.
- ✓ Do not measure from the drawing.

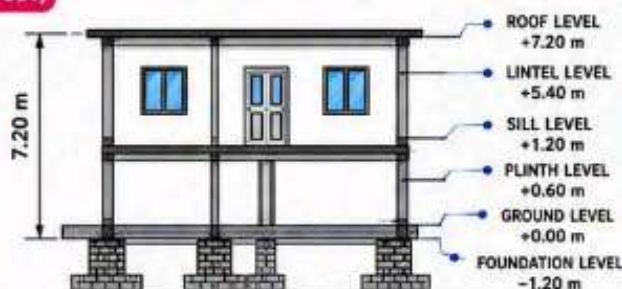
EXAMPLE

4.50 m = 4.50 meters
5.00 m = 5.00 meters

6 SECTION (ELEVATION CUT)

Section shows the vertical cut of the building.

It helps to understand heights and levels.



SECTION A-A

7 ELEVATION (EXTERNAL VIEW)

Elevation shows the external view of the building.

It helps to understand the appearance.



FRONT ELEVATION

8 GENERAL NOTES

- ✓ Read all notes carefully.
- ✓ It contains important instructions.
- ✓ Follow material specifications.
- ✓ Follow workmanship guidelines.



BENEFITS

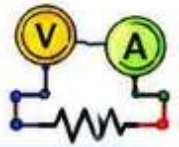
- ✓ Accurate interpretation
- ✓ Better planning & execution
- ✓ Saves time & cost
- ✓ High quality construction





ELECTRICAL FORMULAS

WITH ADVANCED LEVEL CALCULATIONS



1 OHM'S LAW

FORMULA

$$V = I \times R$$

Also,

$$I = \frac{V}{R} \quad R = \frac{V}{I}$$

Units: V (V), I (A), R (Ω)

ADVANCED CALCULATION

Given: $V = 230 \text{ V}$, $R = 46 \Omega$

• Find $I = ?$

$$I = \frac{V}{R} = \frac{230}{46} = 5.00 \text{ A}$$

Given: $V = 48 \text{ V}$, $I = 6 \text{ A}$

• Find $R = ?$

$$R = \frac{V}{I} = \frac{48}{6} = 8 \Omega$$

2 CURRENT

FORMULA

$$I = \frac{V}{R}$$

Also,

$$I = \frac{P}{V}$$

Units: A (Ampere)

ADVANCED CALCULATION

Given: $P = 1200 \text{ W}$, $V = 240 \text{ V}$

• Find $I = ?$

$$I = \frac{P}{V} = \frac{1200}{240} = 5.00 \text{ A}$$

Given: $V = 110 \text{ V}$, $R = 22 \Omega$

• Find $I = ?$

$$I = \frac{V}{R} = \frac{110}{22} = 5.00 \text{ A}$$

3 RESISTANCE

FORMULA

$$R = \frac{V}{I}$$

Also,

$$R = \rho \frac{L}{A}$$

Units: Ω

ADVANCED CALCULATION

Given: $V = 12 \text{ V}$, $I = 0.6 \text{ A}$

• Find $R = ?$

$$R = \frac{V}{I} = \frac{12}{0.6} = 20 \Omega$$

Given: $\rho = 1.72 \times 10^{-8} \Omega\text{m}$, $L = 10 \text{ m}$, $A = 2 \text{ mm}^2$

• Find $R = ?$

$$R = \frac{\rho L}{A} = \frac{1.72 \times 10^{-8} \times 10}{2 \times 10^{-6}} = 0.086 \Omega$$

4 POWER

FORMULAS

$$P = V \times I$$

$$P = I^2 R$$

$$P = \frac{V^2}{R}$$

$$P = \frac{V^2}{R}$$

Units: W (Watt)

ADVANCED CALCULATION

Given: $V = 230 \text{ V}$, $R = 46 \Omega$

• Find $P = ?$

$$P = \frac{V^2}{R} = \frac{230^2}{46} = 1150 \text{ W}$$

Given: $I = 8 \text{ A}$, $R = 10 \Omega$

• Find $P = ?$

$$P = I^2 R = 8^2 \times 10 = 640 \text{ W}$$

Given: $V = 48 \text{ V}$, $I = 6 \text{ A}$

• Find $P = ?$

$$P = V \times I = 48 \times 6 = 288 \text{ W}$$

5 ENERGY

FORMULA

$$E = P \times t$$

Units: J or kWh

ADVANCED CALCULATION

• Given: $P = 1500 \text{ W}$, $t = 3 \text{ h}$

• Find E (kWh) = ?

$$E = P \times t = 1500 \times 3 = 4500 \text{ Wh} = 4.5 \text{ kWh}$$

• Given: $P = 2000 \text{ W}$, $t = 2.5 \text{ h}$

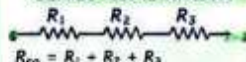
E (J) = $P \times t = 2000 \times (2.5 \times 3600) = 18,000,000 \text{ J}$

USEFUL CONVERSIONS



1 kWh = $3.6 \times 10^6 \text{ J}$
1 HP = 746 W
1 kVA = 1000 VA
1 μF = 10^{-6} F
1 mH = 10^{-3} H

SERIES COMBINATION



Example: $R_1 = 10 \Omega$, $R_2 = 20 \Omega$, $R_3 = 30 \Omega$
 $R_{eq} = 10 + 20 + 30 = 60 \Omega$

PARALLEL COMBINATION



$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$
Example: 10Ω , 20Ω , 30Ω
 $R_{eq} = 5.45 \Omega$

6 AC POWER (SINGLE PHASE)

FORMULA

$$P = VI \cos \theta$$

$$Q = VI \sin \theta$$

$$S = VI$$

Units: W, VAR, VA

ADVANCED CALCULATION

Given: $V = 230 \text{ V}$, $I = 10 \text{ A}$, $\theta = 30^\circ$

• $P = VI \cos \theta = 230 \times 10 \times \cos 30^\circ$

$$= 2300 \times 0.866 = 1981.8 \text{ W}$$

• $Q = VI \sin \theta = 230 \times 10 \times \sin 30^\circ$

$$= 2300 \times 0.5 = 1150 \text{ VAR}$$

$$S = VI = 2300 \text{ VA}$$

7 AC POWER (THREE PHASE)

FORMULAS

$$P = \sqrt{3} V_L I_L \cos \theta$$

$$Q = \sqrt{3} V_L I_L \sin \theta$$

$$S = \sqrt{3} V_L I_L$$

Units: W, VAR, VA

Given: $V_L = 415 \text{ V}$, $I_L = 20 \text{ A}$, $\theta = 30^\circ$

• $P = \sqrt{3} \times 415 \times 20 \times 0.866$

$$= 12441 \text{ W} \quad 12.44 \text{ kW}$$

$Q = \sqrt{3} \times 415 \times 20 \times 0.5$

$$= 7190 \text{ VAR} \quad 7.19 \text{ kVAR}$$

$$S = \sqrt{3} \times 415 \times 20 = 14,360 \text{ VA}$$

8 FREQUENCY

FORMULA

$$f = \frac{1}{T}$$

Also,

$$\omega = 2\pi f$$

Units: Hz, rad/s

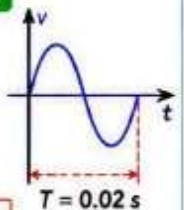
Given: $T = 0.02 \text{ s}$

• Find f and $\omega = ?$

$$f = \frac{1}{T} = \frac{1}{0.02} = 50 \text{ Hz}$$

$$\omega = 2\pi f = 2\pi \times 50$$

$$= 314.16 \text{ rad/s}$$



9 CAPACITANCE

FORMULA

$$Q = C \times V$$

$$X_C = \frac{1}{2\pi f C}$$

Units: F, Ω

Given: $C = 10 \mu\text{F}$, $V = 230 \text{ V}$

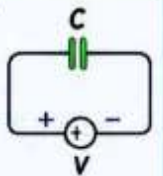
• Find $Q = ?$

$$Q = C \times V = 10 \mu\text{F} \times 230 = 2.30 \times 10^{-3} \text{ C}$$

Given: $f = 50 \text{ Hz}$, $C = 10 \mu\text{F}$

• Find $X_C = ?$

$$X_C = \frac{1}{2\pi f C} = \frac{1}{2\pi \times 50 \times 10^{-5}} = 318.3 \Omega$$



10 INDUCTIVE REACTANCE

FORMULA

$$X_L = 2\pi f L$$

Units: Ω

Given: $f = 50 \text{ Hz}$, $L = 0.1 \text{ H}$

• Find $X_L = ?$

$$X_L = 2\pi \times 50 \times 0.1 = 31.42 \Omega$$



11 CAPACITIVE REACTANCE

FORMULA

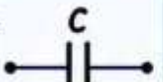
$$X_C = \frac{1}{2\pi f C}$$

Units: Ω

Given: $f = 60 \text{ Hz}$, $C = 20 \mu\text{F}$

• Find $X_C = ?$

$$X_C = \frac{1}{2\pi \times 60 \times 20 \times 10^{-6}} = 132.63 \Omega$$



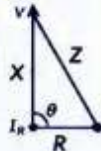
12 IMPEDANCE

FORMULA

$$Z = \sqrt{R^2 + X^2}$$

$$\tan \phi = \frac{X}{R}$$

Units: Ω , rad



ADVANCED CALCULATION

Given: $R = 20 \Omega$, $X_L = 15 \Omega$

• Find Z and $\phi = ?$

$$Z = \sqrt{20^2 + 15^2} = \sqrt{625} = 25 \Omega$$

$$\phi = \tan^{-1} (15/20) = 36.87^\circ$$

Given: $R = 30 \Omega$, $X_C = 25 \Omega$

• Find $Z = ?$

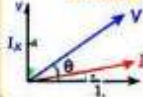
$$Z = \sqrt{30^2 + 25^2} = 39.05 \Omega$$

$$\phi = -\tan^{-1} (25/30) = -39.81^\circ$$

POWER FACTOR

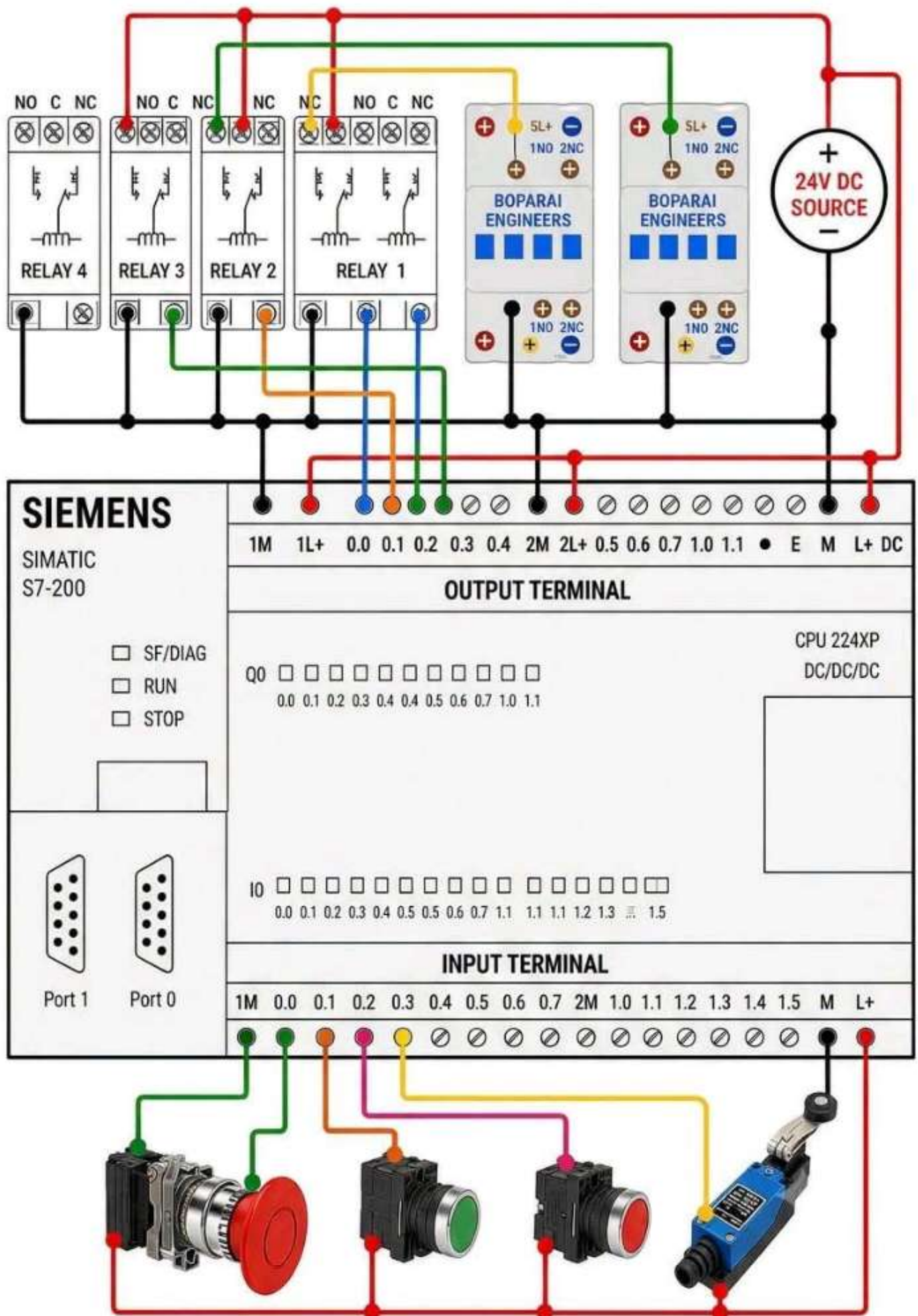
• Lagging: Inductive (I lags V)
• Leading: Capacitive (I leads V)
• Unity PF: $\theta = 0^\circ$, PF = 1

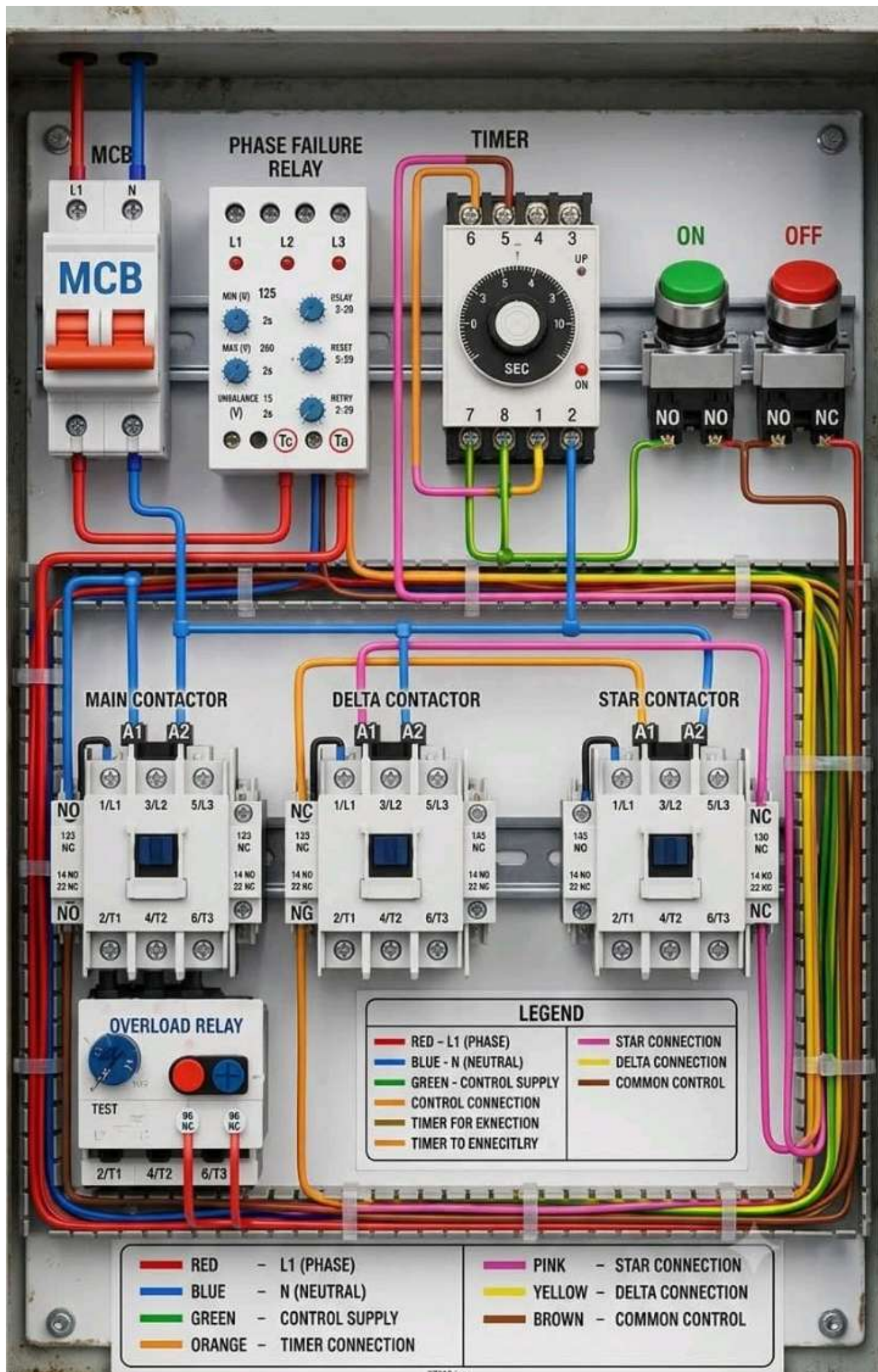
PHASOR RELATIONS



• $P = VI \cos \theta$
• $Q = VI \sin \theta$
• $S = VI$

SIEMENS S7-200 PLC WIRING





Power your Wellness Business with WhatsApp API



Chat.
Convert.



Book.
Confirm.



Grow.
Together.



HR itu **Human Resources** **Bukan** Human Robot

HR bukan robot yang kerjanya cuma input data. Biarkan Gadjian ambil tugas admin anda, semua urusan beres dalam sekali klik.



**Hitung
Gaji**



**Hitung PPH 21
& BPJS**



**Kelola
Absensi**



**Kelola
Inventaris**



**Kelola
Rekrutmen**



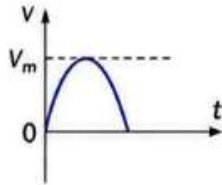
ELECTRICAL CALCULATION



By Electrical Knowledge

1 Average Value of AC Sine Wave

$$V_{avg} = \frac{2 V_m}{\pi}$$



V_{avg} = Average Value (V)
 V_m = Maximum Value (V)
 π = 3.1416

Example:

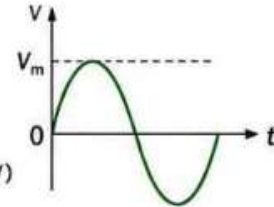
If $V_m = 325$ V

$$V_{avg} = \frac{2 \times 325}{3.1416}$$

$$V_{avg} = 206.6 \text{ V}$$

2 RMS Value of AC Sine Wave

$$V_{rms} = \frac{V_m}{\sqrt{2}}$$



V_{rms} = RMS Value (V)
 V_m = Maximum Value (V)

Example:

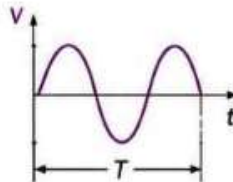
If $V_m = 325$ V

$$V_{rms} = \frac{325}{\sqrt{2}}$$

$$V_{rms} = 229.81 \text{ V}$$

3 Frequency from Time Period

$$f = \frac{1}{T}$$



f = Frequency (Hz)
 T = Time Period (s)

Example:

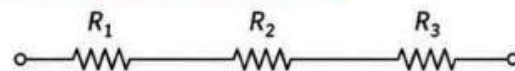
If $T = 0.02$ s

$$f = \frac{1}{0.02}$$

$$f = 50 \text{ Hz}$$

4 Total Resistance in Series Circuit

$$R_T = R_1 + R_2 + R_3 + \dots$$



R_T = Total Resistance (Ω)
 R_i = Individual Resistances (Ω)

Example:

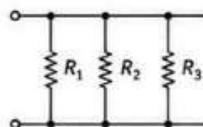
If $R_1 = 4 \Omega$, $R_2 = 6 \Omega$, $R_3 = 10 \Omega$

$$R_T = 4 + 6 + 10$$

$$R_T = 20 \Omega$$

5 Total Resistance in Parallel Circuit

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$$



R_T = Total Resistance (Ω)
 R_i = Individual Resistances (Ω)

Example:

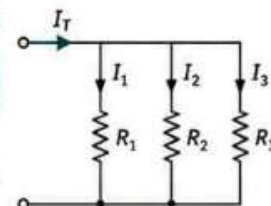
If $R_1 = 6 \Omega$, $R_2 = 3 \Omega$, $R_3 = 2 \Omega$

$$\frac{1}{R_T} = \frac{1}{6} + \frac{1}{3} + \frac{1}{2} = \frac{1+2+3}{6} = 1$$

$$R_T = \frac{1}{1} = R_T = 1 \Omega$$

6 Current in a Parallel Circuit

$$I_T = I_1 + I_2 + I_3 + \dots$$



I_T = Total Current (A)
 I_i = Branch Currents (A)

Example:

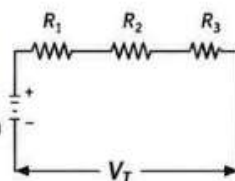
If $I_1 = 2$ A, $I_2 = 3$ A, $I_3 = 5$ A

$$I_T = 2 + 3 + 5$$

$$I_T = 10 \text{ A}$$

7 Voltage in a Series Circuit

$$V_T = V_1 + V_2 + V_3 + \dots$$



V_T = Total Voltage (V)
 V_i = Voltage Across Resistors (V)

Example:

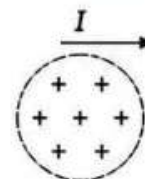
If $V_1 = 20$ V, $V_2 = 30$ V, $V_3 = 50$ V

$$V_T = 20 + 30 + 50$$

$$V_T = 100 \text{ V}$$

8 Electric Charge

$$Q = I \times t$$



Q = Charge (Coulomb)
 I = Current (A)
 t = Time (s)

Example:

If $I = 5$ A, $t = 120$ s

$$Q = 5 \times 120$$

$$Q = 600 \text{ C}$$

ELECTRICAL CALCULATION



By Electrical Knowledge

1 Power from Current & Resistance

$$I = 10 \text{ A}$$

$$R = 20 \Omega$$

$$P = ?$$



$$P = I^2 \times R$$

$$P = 10^2 \times 20$$

$$P = 100 \times 20$$

$$P = 2000 \text{ W}$$

$$P = 2000 \text{ W}$$

2 Current from Power & Voltage

$$P = 3 \text{ kW}$$

$$V = 240 \text{ V}$$

$$I = ?$$



$$I = P / V$$

$$I = 3000 / 240$$

$$I = 12.5 \text{ A}$$

$$I = 12.5 \text{ A}$$

3 Resistance from Voltage & Current

$$V = 360 \text{ V}$$

$$I = 18 \text{ A}$$

$$R = ?$$



$$R = V / I$$

$$R = 360 / 18$$

$$R = 20 \Omega$$

$$R = 20 \Omega$$

4 Power Factor Calculation

$$P = 8 \text{ kW}$$

$$S = 10 \text{ kVA}$$

$$PF = ?$$



$$PF = P / S$$

$$PF = 8 / 10$$

$$PF = 0.80$$

$$PF = 0.80$$

5 Apparent Power (kVA) Calculation

$$P = 6 \text{ kW}$$

$$PF = 0.75$$

$$S = ?$$



$$S = P / PF$$

$$S = 6 / 0.75$$

$$S = 8 \text{ kVA}$$

$$S = 8 \text{ kVA}$$

6 Reactive Power (kVAR) Calculation

$$S = 10 \text{ kVA}$$

$$P = 8 \text{ kW}$$

$$Q = ?$$



$$Q = \sqrt{S^2 - P^2}$$

$$Q = \sqrt{10^2 - 8^2}$$

$$Q = \sqrt{100 - 64}$$

$$Q = \sqrt{36} = 6 \text{ kVAR}$$

$$Q = 6 \text{ kVAR}$$

7 Energy Consumption (kWh) Calculation

$$P = 4.5 \text{ kW}$$

$$t = 8 \text{ h}$$

$$E = ?$$



$$E = P \times t$$

$$E = 4.5 \times 8$$

$$E = 36 \text{ kWh}$$

$$E = 36 \text{ kWh}$$

8 Three Phase Line Current Calculation

$$P = 18 \text{ kW}$$

$$V_L = 415 \text{ V}$$

$$PF = 0.85$$

$$I_L = ?$$



$$P = \sqrt{3} \times V_L \times I_L \times PF$$

$$I_L = P / (\sqrt{3} \times V_L \times PF)$$

$$I_L = \frac{18000}{(1.732 \times 415 \times 0.85)}$$

$$I_L = 29.45 \text{ A}$$

$$I_L = 29.45 \text{ A}$$

NEW APP

AI POWERED

AUTOMATED TRADING BOT

Smart technology.

Runs 24/7.

You stay in control.



FULLY AUTOMATED

The bot analyzes the market and executes trades for you.



DATA-DRIVEN STRATEGY

Advanced algorithms find high-potential opportunities.



RISK AWARE

Built-in protection systems help manage market risks.



FAST & RELIABLE

Trades executed in real time with speed and accuracy.



SAFE & SECURE

Top-tier security keeps your data and funds protected.



OPEN CHAT • 100% TRANSPARENCY

- Live communication with our team
- See real results in real time
- No hidden fees. No secrets.

LIVE



TRY THE BOT TODAY

Set it up once and let it work.



Business logic vulnerability



Watch

163



Star

1,018

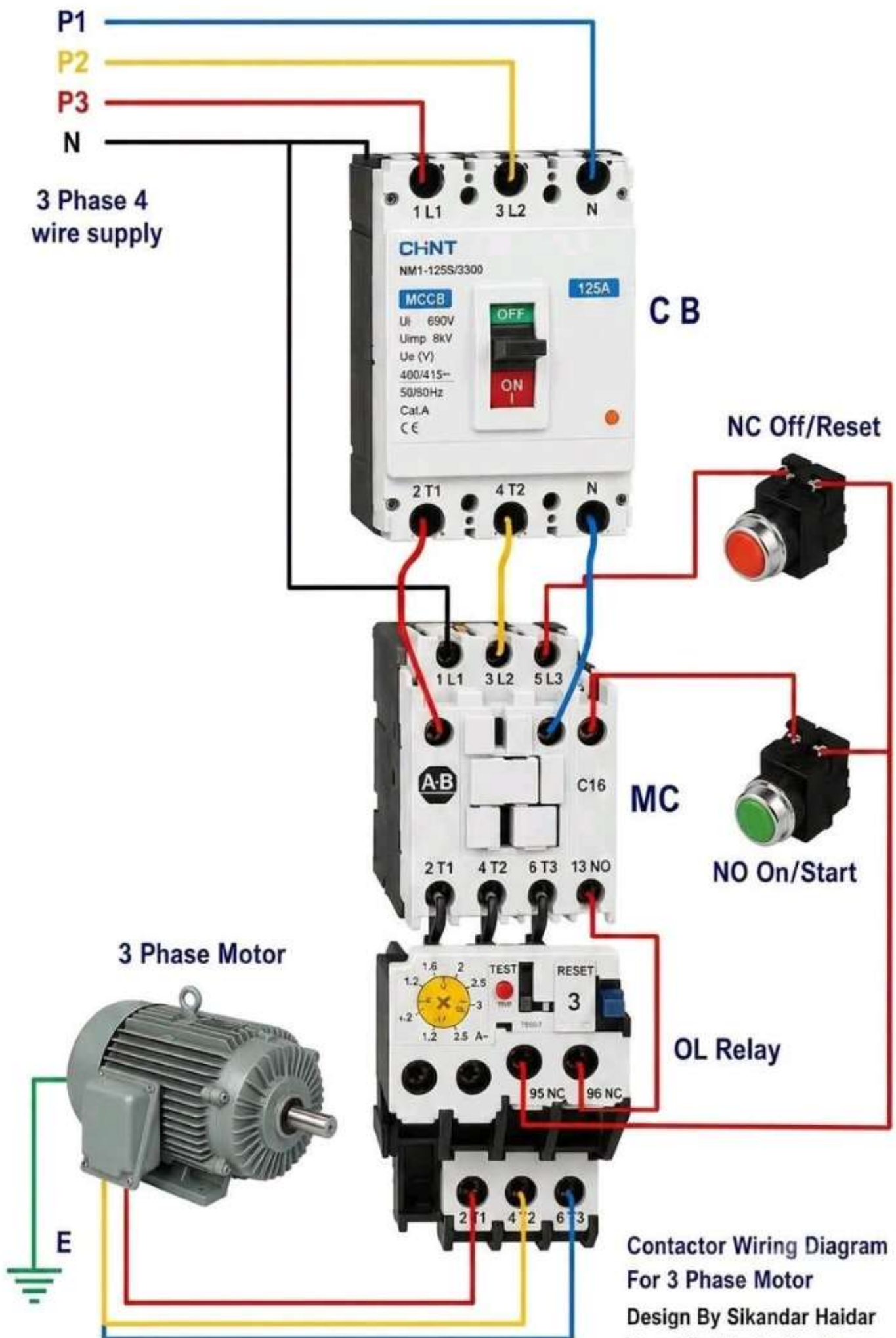
NVD Categorization

CWE-840: Business Logic Errors:

Weaknesses in this category identify some of the underlying problems that commonly allow attackers to manipulate the business logic of an application. Errors in business logic can be devastating to an entire application. They can be difficult to find automatically, since they typically involve legitimate use of the

apa itu bug yang namanya bisnis logic

are similar to well-understood implementation and design weaknesses



**Contactor Wiring Diagram
For 3 Phase Motor**
Design By Sikandar Haidar
From Electricalonline4u.com

Errors Affecting Chainage (Slope, Temperature, Sag, Tension)

1. SLOPE ERROR



Formula

$$L_h = L_s \cos \theta$$

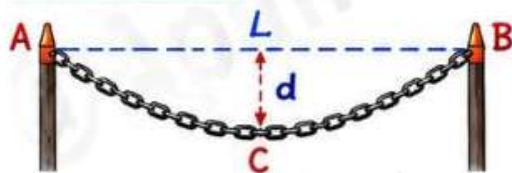
Error (due to slope) =

$$L_s - L_h = L_s (1 - \cos \theta)$$

Where:

- L_s = Measured length along slope
- L_h = True horizontal length
- θ = Angle of slope

3. SAG ERROR



Length measured along chain = L_m

True horizontal length = L_h

Formula:

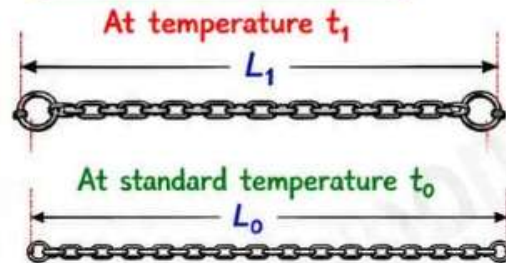
$$d = \frac{w L^2}{8P}$$

Where:

- w = Weight of chain per unit length
- P = Pull or tension in the chain
- L = Horizontal length

$$\text{Correction for sag } (C_s) = \frac{w L^3}{24 P^2}$$

2. TEMPERATURE ERROR



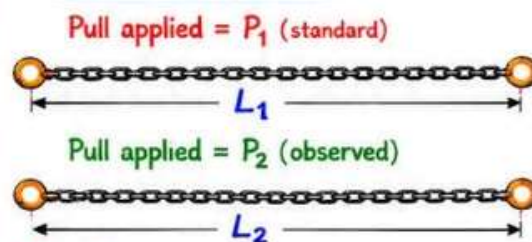
Formula:

$$\Delta L_T = \alpha L_0 (t_1 - t_0)$$

Where:

- α = Coefficient of linear expansion (per °C)
- L_0 = Standard length of the chain
- t_1 = Observed temperature
- t_0 = Standard temperature (usually 20°C)

4. TENSION ERROR



Formula

$$L_2 = L_1 \frac{P_2}{P_1}$$

Error due to tension =

$$L_2 - L_1$$

Where:

- P_1 = Standard pull
- P_2 = Observed pull
- L_1 = Standard length
- L_2 = Corrected length

COMPREHENSIVE CORRECTION OF CHAINAGE



$$L_{\text{corrected}} = L_{\text{measured}} \pm C_{\text{slope}} \pm C_{\text{temp}} \pm C_{\text{sag}} \pm C_{\text{tension}}$$

(Slope Correction)
(Temperature Correction)
(Sag Correction)
(Tension Correction)

QUICK TIPS

- ✓ Always apply corrections before final computation of chainage.
- ✓ Use standard chain length = 20m (unless otherwise stated).
- ✓ Keep chain straight and level to minimize errors.



REMEMBER!

- ✓ Slope → use $\cos \theta$
- ✓ Temperature → use $\alpha L_0 (t_1 - t_{00})$
- ✓ Sag → increases with length, decreases with tension
- ✓ Tension → higher pull = longer chain

2Ah • 4Ah 5Ah • 8Ah APA BEDANYA?

Perbedaan utamanya adalah kapasitas energi,
bukan sekadar tenaga.



YANG UMUM TERJADI



2Ah

- ✓ Lebih ringan
- ✓ Cocok pekerjaan singkat



4Ah-5Ah

- ✓ Seimbang
- ✓ Banyak dipilih untuk penggunaan harian



8Ah

- ✓ Durasi kerja lebih lama
- ✓ Lebih berat



MITOS

8Ah pasti lebih bertenaga daripada 2Ah.



VS



FAKTA

Nilai Ah menunjukkan kapasitas baterai. Lama penggunaan biasanya meningkat, sedangkan performa alat dipengaruhi oleh desain mesin dan sistem baterainya.



Pekerjaan ringan → baterai kecil lebih nyaman



Pekerjaan lama → kapasitas lebih besar mengurangi frekuensi pengisian



Pertimbangkan keseimbangan antara durasi kerja dan bobot alat

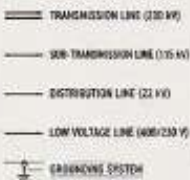


Kapasitas yang lebih besar umumnya membuat baterai lebih berat. Pilih sesuai kebutuhan, bukan hanya angka terbesar.

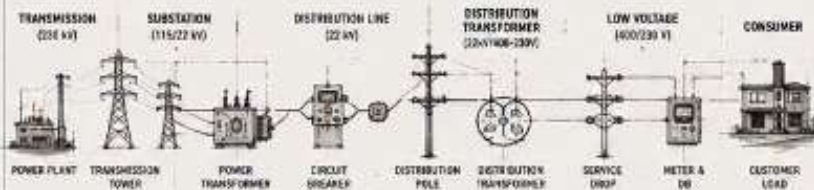
POWER DISTRIBUTION SYSTEM



LEGEND



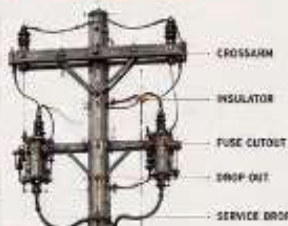
SINGLE LINE DIAGRAM - DISTRIBUTION SYSTEM



ELECTRIC TECHNICIAN

POWERING SYSTEMS. ENSURING SAFETY.

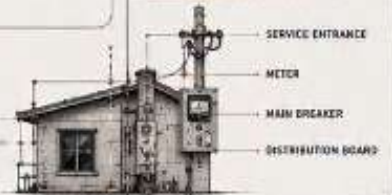
DISTRIBUTION POLE (22 kV)



DISTRIBUTION TRANSFORMER



LOW VOLTAGE SERVICE



DISTRIBUTION BOARD



INSTRUMENT & TOOLS



SAFETY FIRST





Integrated Data Center Infrastructure

Built for AI, edge, and
high-density computing.



Power



Cooling



IT Space



Monitoring



Fire Protection



Security

Contact Us



TYPES OF SUPPORT CONNECTIONS

In structures, the way a member is connected to its support defines how it can move or resist forces.

FIXED SUPPORT

No translation
No rotation

Resists all forces and moments.

- ☐ Resists F_x ($\rightarrow$)
- ☐ Resists F_y ($\uparrow$)
- ☐ Resists M ($\curvearrowright$)



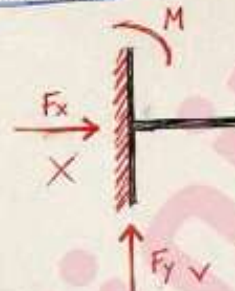
Column base with anchor bolts on concrete



Beam fixed to concrete wall with end plate connection



SCHEMATIC



FIXED

No Translation
No Rotation

Example:
Building columns,
Rigid beam connections,
Base plates.

PINNED SUPPORT

Allows rotation
No translation (perpendicular)

Resists forces but not moment.

- ☐ Resists F_x ($\rightarrow$)
- ☐ Resists F_y ($\uparrow$)
- ☐ Does NOT resist M ($\curvearrowright$)

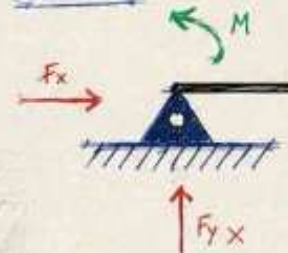


Pinned base with clevis and pin connection



Pinned connection using a clevis and pin

SCHEMATIC



PINNED

Allows Rotation $\checkmark$

No Vertical Translation

Example:
Truss joints, Hinged beams,
Machine supports.

ADDITIONAL NOTES

• Real connections are idealizations.

• Design depends on function & behavior.



ROLLER SUPPORT

Allows rotation
Allows translation (horizontal)

Resists only one force (perpendicular).

- ☐ Resists F_y ($\uparrow$)
- ☐ Does NOT resist F_x ($\rightarrow$)
- ☐ Does NOT resist M ($\curvearrowright$)

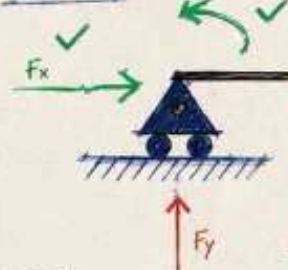


Bridge support with roller bearings



Roller support under a heavy structure

SCHEMATIC



ROLLER

Allows Rotation $\checkmark$
Allows Horizontal Translation $\checkmark$

No Vertical Translation

Example:
Bridges, Conveyors,
Sliding structures.

QUICK SUMMARY

FIXED
No movement
Resists F_x , F_y , M

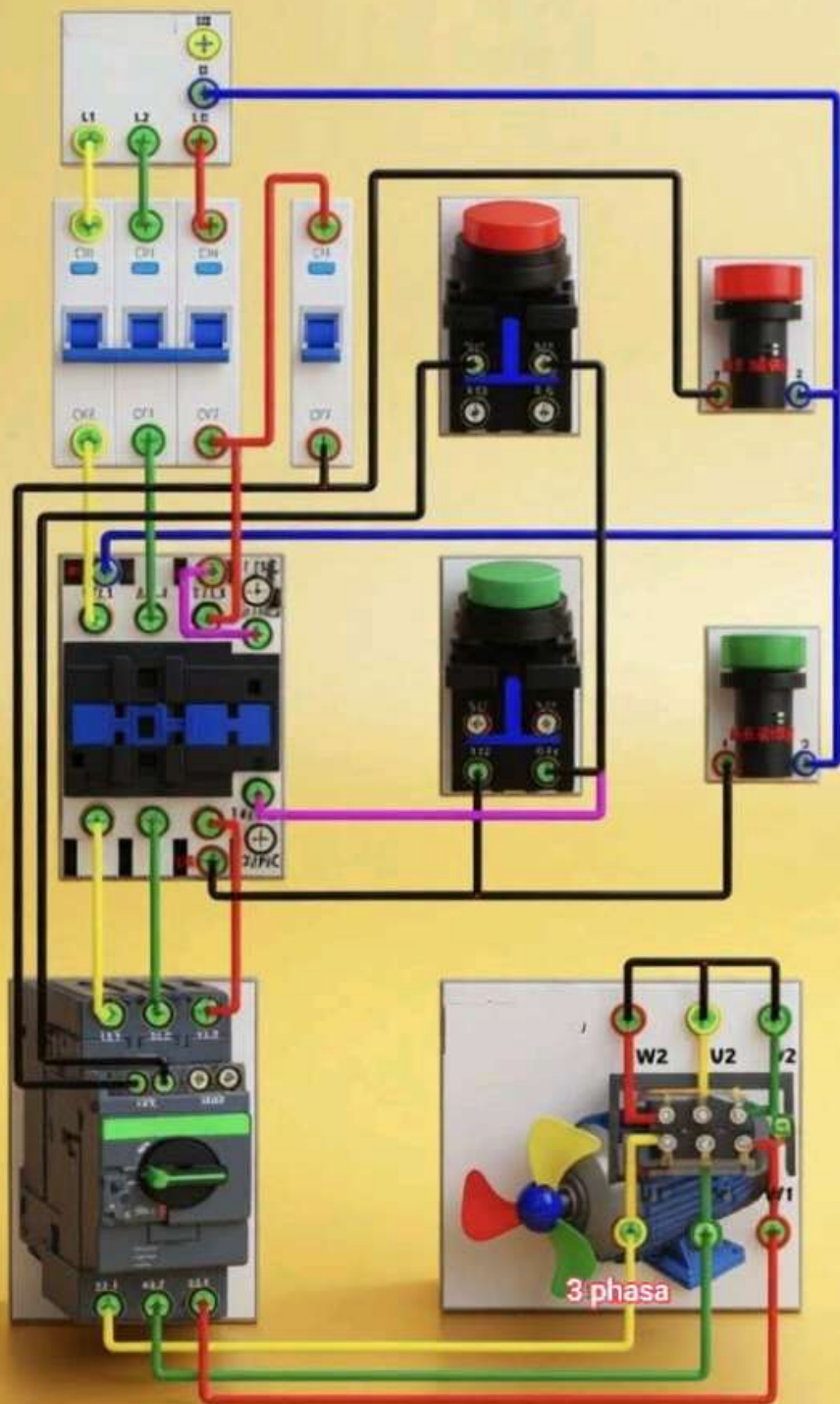
PINNED
Allows rotation
Resists F_x , F_y

ROLLER
Allows rotation + horizontal movement
Resists F_y only

REMEMBER!

The right support connection ensures stability, safety, and proper behavior of the structure.





VEEAM EXCLUSIVE

veeam

Hybrid Cloud Backup E-Book

Mastering cost control
and security

LEARN MORE



JASA INSTALL ULANG & SERVIS LAPTOP / PC PANGGILAN

SAMARINDA

LAPTOP LEMOT? SUSAH KERJA? JANGAN PANIK!

SERVICES MELIPUTI:

-  INSTALASI ULANG WINDOWS
(XP, 7, 8.1, 10, 11)
-  OPTIMASI & PERCEPATAN
PC/LAPTOP
-  INSTALASI SOFTWARE
(Office, Adobe, Corel, AutoCAD, dll)
-  PEMBERSIHAN
VIRUS &
-  PEMBERSIHAN VIRUS &
TROUBLESHOOTING

BERPENGALAMAN SEJAK 2012

✓ AHLI, CEPAT, & TERPERCAYA

HUBUNGI SAYA SEKARANG!



+62 823 5033 1742

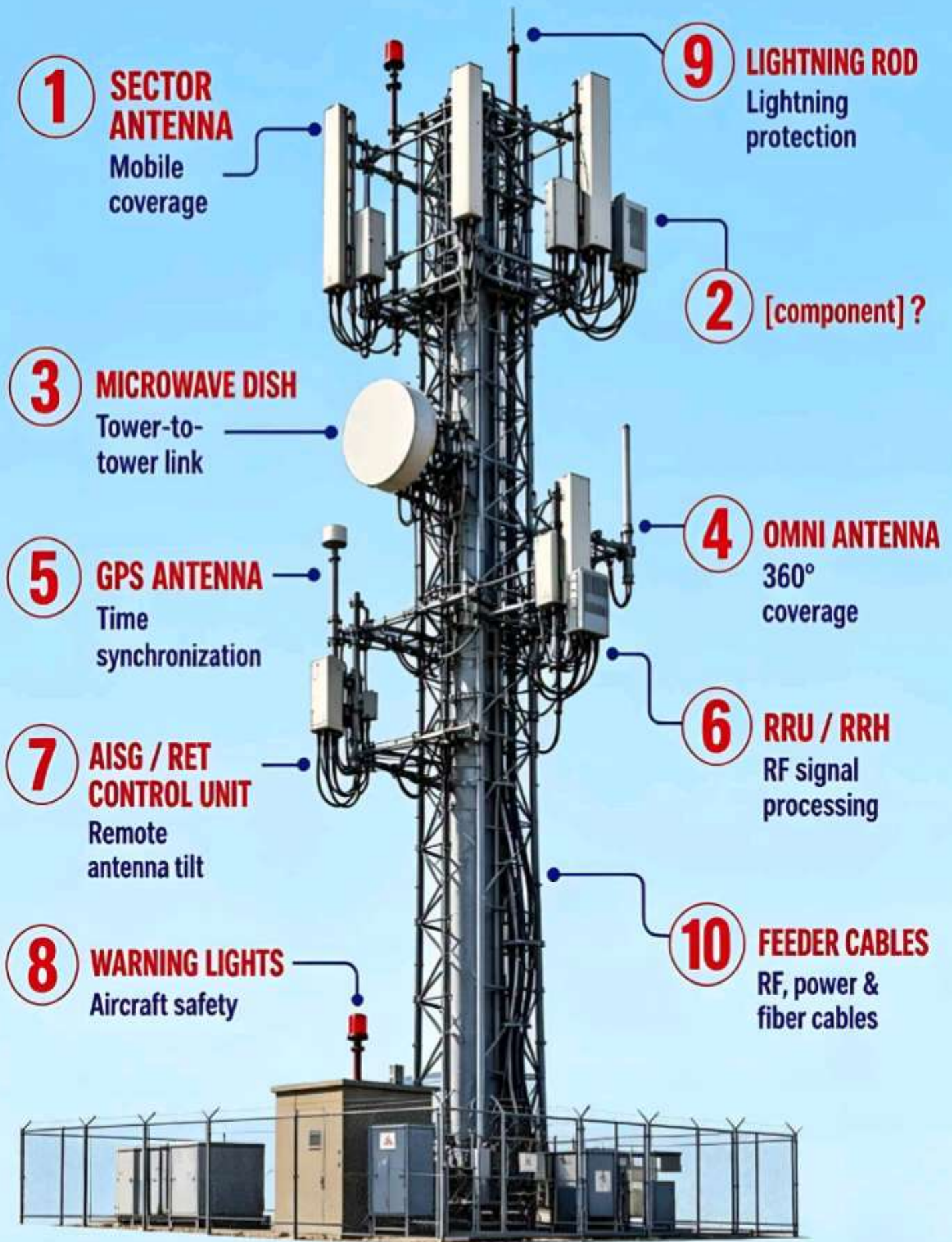
(Call / SMS / WhatsApp)

LAYANAN PANGGILAN KE RUMAH / KANTOR



Samarinda

4G/5G COMMUNICATIONS TOWER



ELECTRICAL FORMULAS

1. Ohm's Law

$$V = I \times R$$

V = Voltage (V)

I = Current (A)

R = Resistance (Ω)

2. Current

$$I = \frac{V}{R}$$

3. Resistance

$$R = \frac{V}{I}$$

4. Power

$$P = V \times I$$

$$P = I^2 R$$

$$P = \frac{V^2}{R}$$

P = Power (W)

5. Energy

$$E = P \times t$$

E = Energy (J or kWh)

P = Power (W)

t = Time (s or h)

Note:

* 1 kWh = 3.6×10^6 J

* 1 HP = 746 W

6. AC Power (Single Phase)

$$P = V \times I \times \cos \theta$$

θ = Phase Angle

7. AC Power (Three Phase)

$$P = \sqrt{3} \times V \times I \times \cos \theta$$

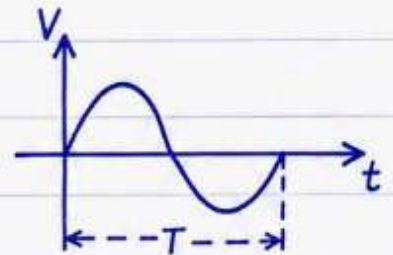
$\sqrt{3} = 1.732$

8. Frequency

$$f = \frac{1}{T}$$

f = Frequency (Hz)

T = Time Period (s)



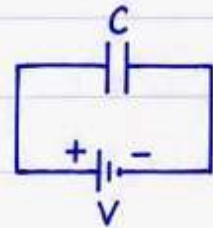
9. Capacitance

$$Q = C \times V$$

Q = Charge (C)

C = Capacitance (F)

V = Voltage (V)



10. Inductive Reactance

$$X_L = 2\pi f L$$

X_L = Inductive Reactance (Ω)

f = Frequency (Hz)

L = Inductance (H)



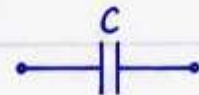
11. Capacitive Reactance

$$X_C = \frac{1}{2\pi f C}$$

X_C = Capacitive Reactance (Ω)

f = Frequency (Hz)

C = Capacitance (F)



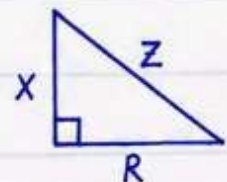
12. Impedance

$$Z = \sqrt{R^2 + X^2}$$

Z = Impedance (Ω)

R = Resistance (Ω)

X = Reactance (Ω)

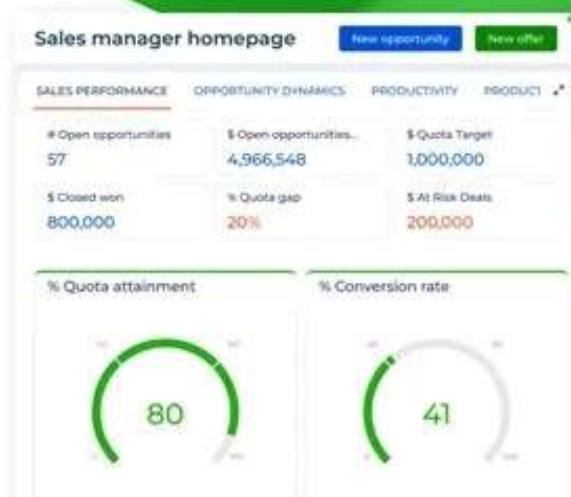




TRUE NO-CODE AI CRM

AI agents that handle routine
tasks for you

Get a Demo



Q4 Forecast Review

How do I affect the sales forecast for this quarter?

Forecast Agent

Focus on closing high-potential opportunities:

The "Contracting" and "Awaiting Sale" stages have significant drop-offs but represent deals close to closing.

Actions:

1. Identify high-value opportunities in these stages and prioritize them.
2. Offer time-sensitive incentives or discounts to encourage decision-making.

Re-engage leads in the pipeline:

There are 85 leads in the pipeline, a large pool for nurturing into opportunities.

Actions:

1. Follow up with inactive leads and rekindle interest. It could bring a +20% conversion rate.
2. Focus on leads that align with your ideal customer profile (ICP) for quicker qualification.

View Full Forecast Adjust Forecasts





VIRTUAL HANDS ON LAB

AGENTIC AI IMMERSION WORKSHOP

30 July

9:30 AM IST | 12:00 PM SGT

2:00 PM AEST

REGISTER NOW

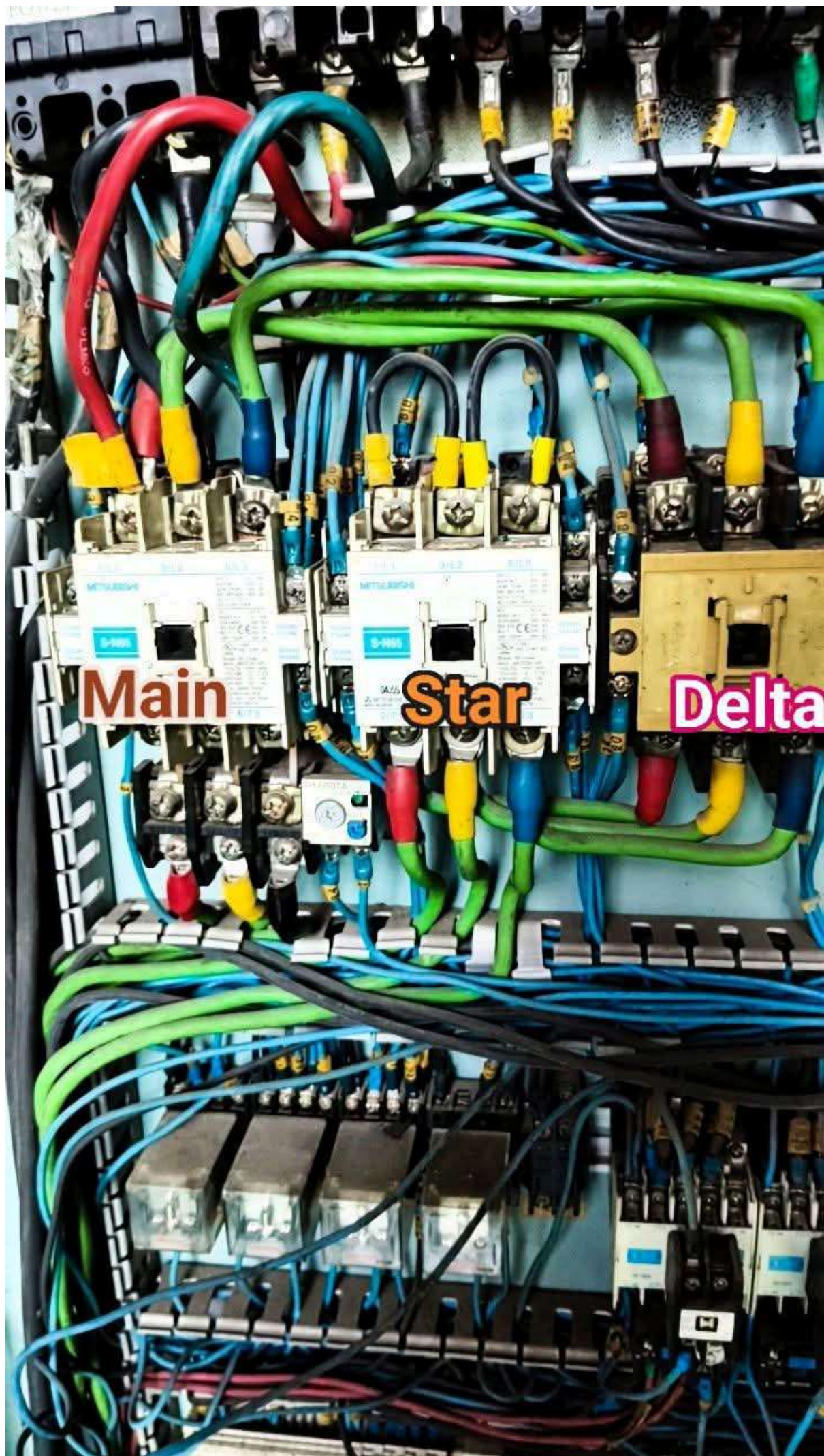


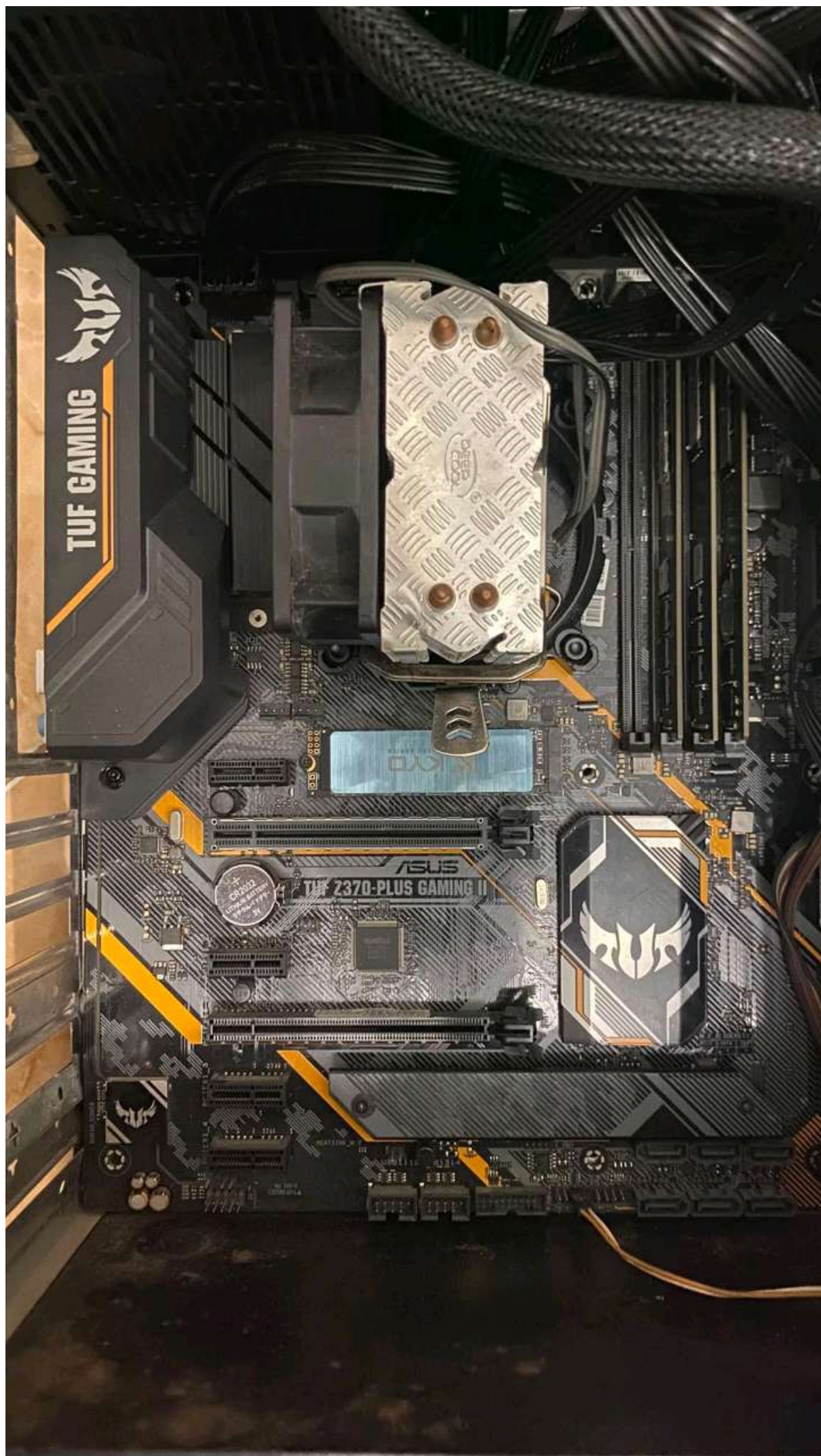
SHEENA NASIM

Principal Architect, Machine Learning
Field CTO APJ
Snowflake

LIVE &
LOCAL









Aivastark

AI AGENT FOR YOUR BUSINESS

ENGAGE • SUPPORT • CONVERT

Turn visitors into customers
with **Smart AI** Conversations



Instant Answers
Happy Customers



Boost Sales
More Conversions



Capture Leads
Grow Your Business



Save Time & Costs
Automate Support

 Facebook

 Whatsapp

 Instagram

 Website

24/7 AI Support That Works
While You Sleep!

24/7



— One AI Agent. Unlimited Possibilities. —



Website Chat



Knowledge Base



Files & Docs



Multi-language



Analytics



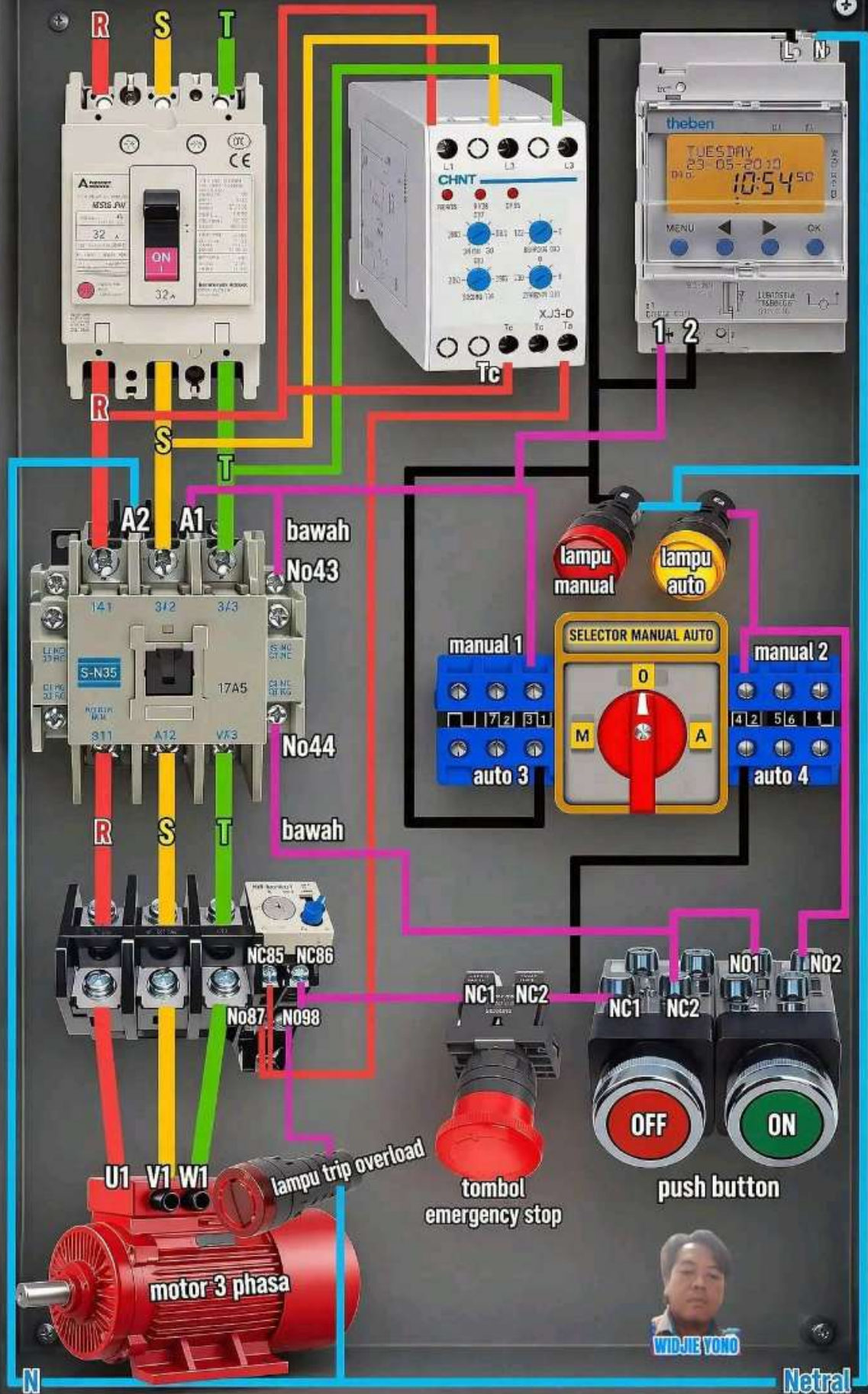
Custom Actions

Smarter Conversations
Stronger Business

GET STARTED TODAY

www.AivaStark.com

DIAGRAM PANEL KONTROL MOTOR 3 FASA DENGAN MANUAL - AUTO & PROTEKSI OVERLOAD



KETERANGAN:

R : Fase R
S : Fase S
T : Fase T
N : Netral

A1-A2 : Coil Kontaktor
NO : Normally Open
NC : Normally Closed
Tc : Timer Close
Ta : Timer Open



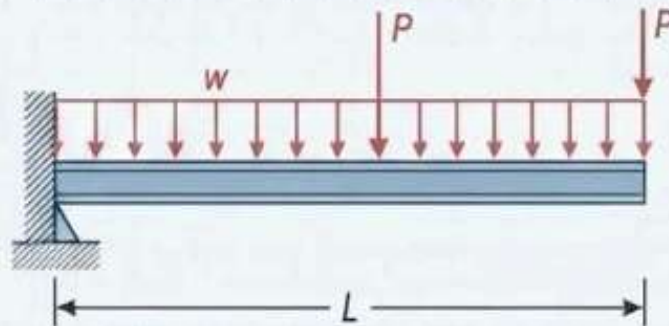
PERHATIAN!

Pastikan seluruh sambungan kencang dan sesuai standar keselamatan kelistrikan.

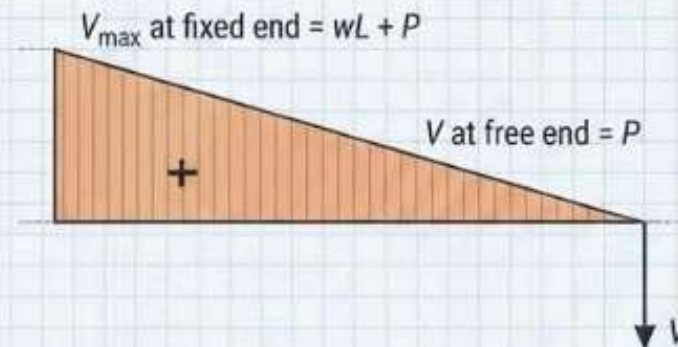


CANTILEVER BEAM WITH UNIFORM LOAD AND POINT LOAD

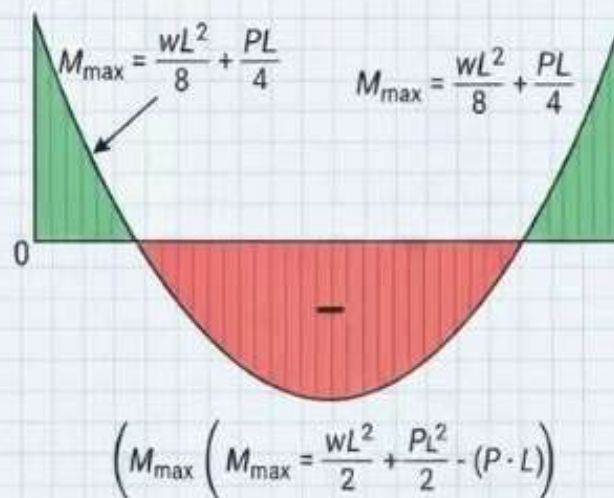
▣ Cantilever beam with length $L=5000$ mm, $w=20$ kN/m, $P=30$ kN ▣



Shear Force Diagram (SFD)



Bending Moment Diagram (BMD)



Calculations

Support reactions

$$R_A = -\frac{w_{\max}}{8} = \frac{PL}{4}$$

$$R = 4200.1518 \text{ PI}$$

$$V_A = \frac{-500 \text{ nx}}{100}$$

$$R_B = \frac{0.33426}{\frac{d}{1} V_{\max}}$$

$$R_B = \frac{-20.0}{-129.00}$$

Maximum Shear

$$M_{\max} = \frac{wL^2}{8} + PL = -\frac{wL^2}{4}$$

$$= \frac{wL^2}{8} + wL + PL$$

Maximum Moment

$$M_{\max} = \frac{wL^2}{8} + \frac{PL}{4}$$

$$M_{\max} = \frac{wL^2}{8} + \frac{PL}{4}$$

$$= 1.653 \text{ mL}$$

$$R_B = \frac{wL^2}{8} + PL$$

Maximum Moment

$$R_A = \frac{wL^2}{8} + \frac{PL}{4}$$

$$= \frac{wL^2}{8} + \frac{PL}{4}$$

$$= 1.2954$$

$$R_A = -0.1556$$

Maximum moment mom

$$M_{\max} = \frac{wL^2}{8} + \frac{wL^2}{10} + \frac{PL}{4}$$

$$= \frac{wL^2}{8} + \frac{PL}{4}$$

$$M_m = 100.0900$$

Beam Properties and Material Data

Young's Modulus

$$E = 200 \text{ GPa}$$

$$E = 200 \text{ GPa}$$

$$I = 120 \times 10^6 \text{ mm}^4$$

Cross section

Cross section

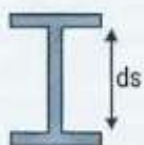
Flange width = 200 mm

Flange thickness = 15 mm

Web thickness = 10 mm

Overall depth = 400 mm

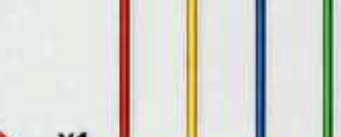
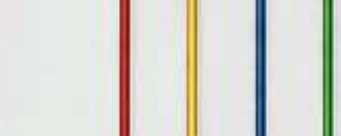
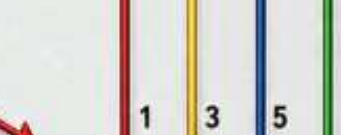
Cross section





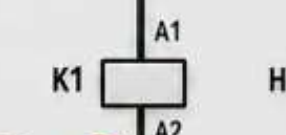
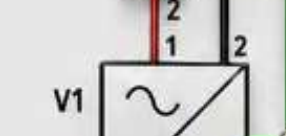
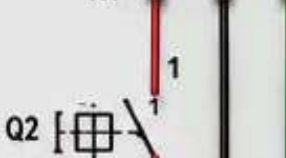
440Vca 3F

L1 L2 L3 PE



220Vca F+N+PE

L N PE



Disjuntor Monopolar

Fonte Retificadora

Botão Emergência

Fusível

Pulsador Desliga

Pulsador Liga

Sinaleiro

Bobina Contator

Disjuntor Motor

Contator de Força

Cabos

Bornes

Motor Trifásico

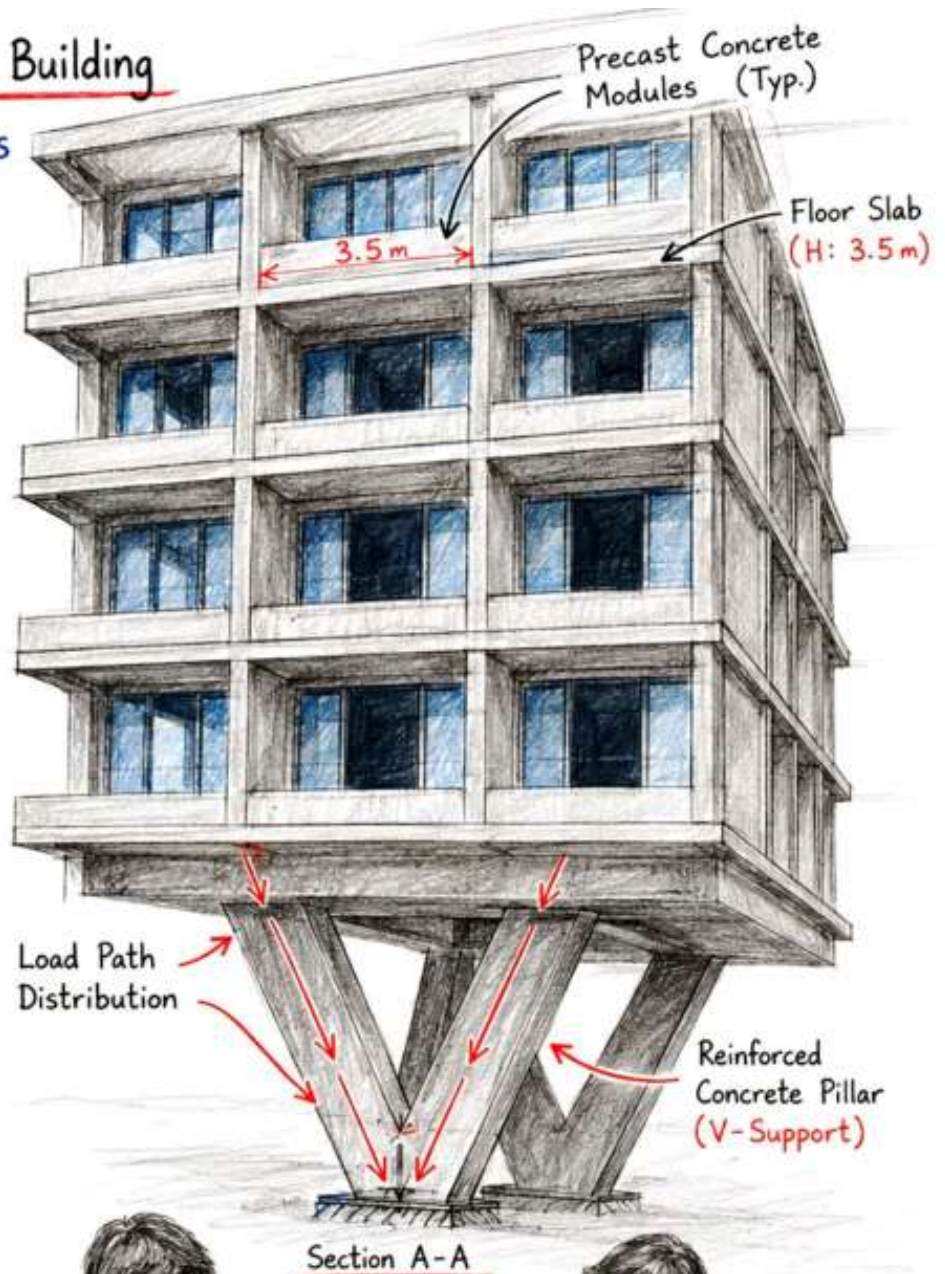
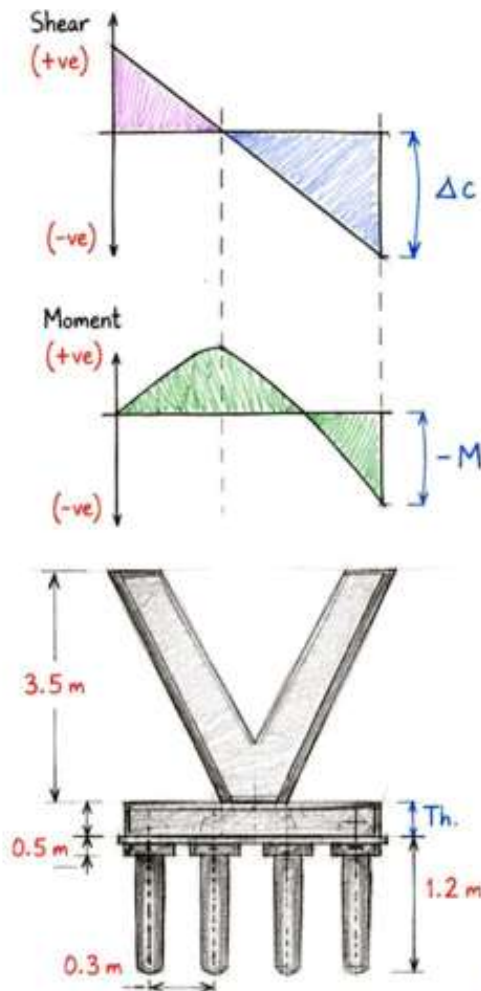


Precast Concrete Modular Building

Structural Concept & Analysis

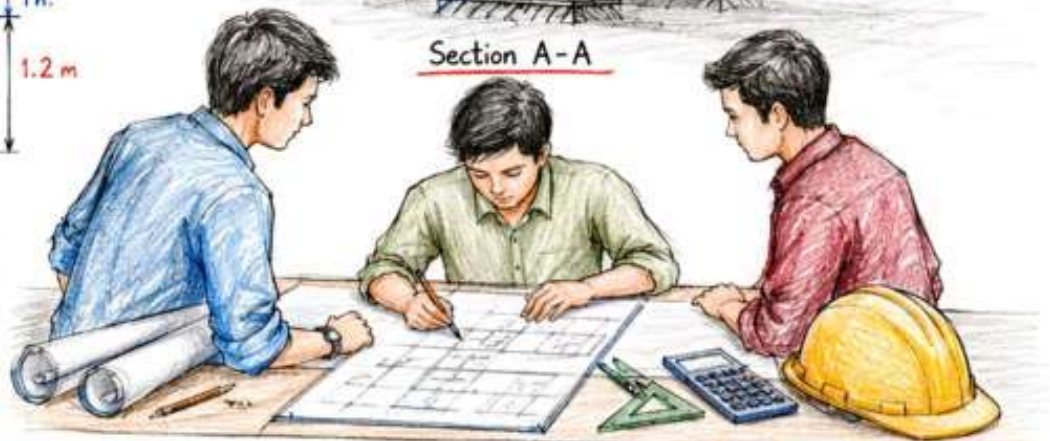
Structural Analysis

Shear & Moment Diagram

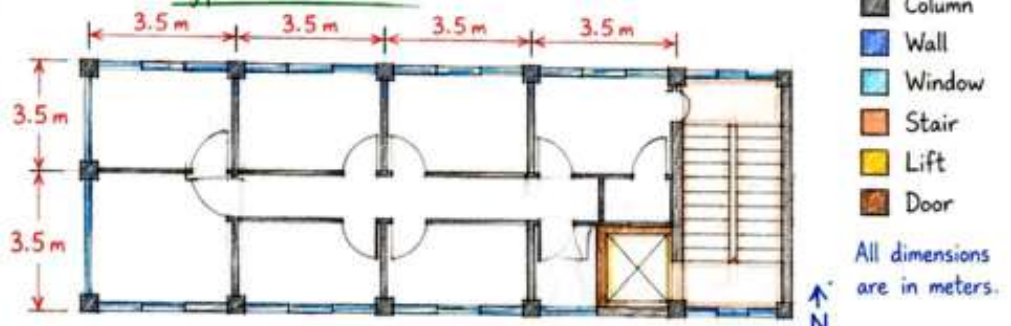


Key Notes

- Modular spacing (typ): 3.5 m
- Floor slab height: 3.5 m
- V-support transfers vertical loads efficiently
- Shear & moment diagrams shown for structural understanding



Typical Floor Plan



SURVEYING CLASS Q&A

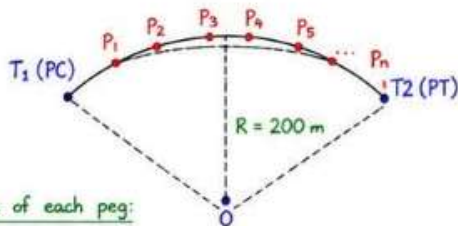
PEG INTERVAL COMPUTATION

Date: 9/7/26

Peg interval is the distance between two consecutive pegs along a curve.

It is used to set out points on a curve at regular intervals.

- Q1.** A simple circular curve has a radius of 200 m. The chainage of T1 (PC) is 10+000 and the chainage of T2 (PT) is 10+314.16. If the peg interval is 20 m, compute the number of pegs required and the chainage of each peg from PC to PT.



Chainage of each peg:

Peg	PC	P ₁	P ₂	P ₃	P ₄	P ₅	P ₆	P ₇	P ₈	P ₉	P ₁₀	P ₁₁	P ₁₃	P ₁₄	P ₁₅	PT
Chainage	10+000	10+020	10+040	10+060	10+080	10+120	10+140	10+160	10+180	10+200	10+220	10+240	10+260	10+280	10+300	10+314.16

Solution:

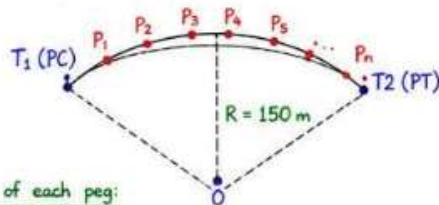
$$\text{Length of Curve, } L = 10+314.16 - 10+000 = 314.16 \text{ m}$$

$$\text{Peg Interval, } d = 20 \text{ m}$$

$$\text{Number of intervals, } n = \frac{L}{d} = \frac{314.16}{20} = 15.708 \approx 16 \text{ intervals}$$

$$\begin{aligned} \text{Number of pegs} &= 16 + 1 \text{ (including PC)} \\ &= 17 \text{ pegs} \end{aligned}$$

- Q2.** A simple circular curve has radius 150 m. The length of the curve is 235.62 m. If the peg interval is 15 m, compute the number of pegs required and the chainage of each peg from PC to PT.



Chainage of each peg:

Peg	PC	P ₁	P ₂	P ₃	P ₅	P ₆	P ₇	P ₈	P ₉	P ₁₀	P ₁₁	P ₁₂	P ₁₃	P ₁₄	P ₁₅	PT
Chainage	10+000	10+015	10+030	10+045	10+075	10+090	10+105	10+120	10+135	10+150	10+165	10+180	10+195	10+210	10+225	10+235.62

Solution:

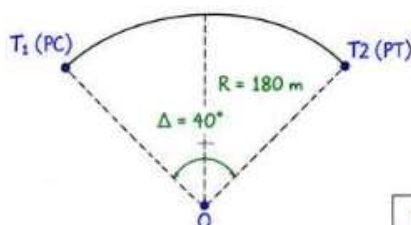
$$\text{Length of Curve, } L = 235.62 \text{ m}$$

$$\text{Peg Interval, } d = 15 \text{ m}$$

$$\text{Number of intervals, } n = \frac{L}{d} = \frac{235.62}{15} = 15.708 \approx 16 \text{ intervals}$$

$$\text{Number of pegs} = 16 + 1 = 17 \text{ pegs}$$

- Q3.** A circular curve has radius 180 m and central angle 40°. If the peg interval is 20 m, calculate the number of pegs required.



Solution:

$$\begin{aligned} \text{Step 1: Length of curve, } L &= \frac{\pi R \Delta}{180} \\ &= \frac{3.142 \times 180 \times 40}{180} \\ &= 125.66 \text{ m} \end{aligned}$$

$$\begin{aligned} \text{Step 2: Number of intervals, } n &= \frac{L}{d} = \frac{125.66}{20} \\ &= 6.283 \approx 6 \text{ intervals} \end{aligned}$$

$$\text{Number of pegs} = 6 + 1 = 7 \text{ pegs}$$

Step 3: Chainage of each peg (assume PC = 5+000)

Peg	PC	P ₁	P ₂	P ₃	P ₄	P ₅	P ₆	PT
Chainage	5+000	5+020	5+040	5+060	5+080	5+100	5+120	5+125.66

KEY FORMULAS

- ✓ Length of Curve $L = \frac{\pi R \Delta}{180}$ (Δ in degrees)
 - ✓ Number of Intervals $n = L/d$
 - ✓ Number of Pegs $= n + 1$ (including PC)
- Where: R = Radius, Δ = Central Angle, d = Peg Interval

NOTE:

- ✓ Always carry the last peg at PT exactly on the tangent.
- ✓ If L is not a multiple of d , the last interval will be shorter.
- ✓ Use consistent chainage and units.

ELECTRICAL CALCULATION

By Electrical Knowledge



1 Voltage Drop in a Conductor

$$I = 20 \text{ A}$$

$$R = 0.35 \text{ } \Omega$$

$$L = 50 \text{ m}$$

$$V_D = ?$$

$$V_D = I \times R \times L$$

$$V_D = 20 \times 0.35 \times 50$$

$$V_D = 350 \text{ V}$$

$$V_D = 350 \text{ V}$$

2 Current through a Conductor

$$P = 9 \text{ kW}$$

$$V = 415 \text{ V}$$

$$\text{PF} = 0.85$$

$$I = ?$$

$$P = \sqrt{3} \times V \times I \times \text{PF}$$

$$I = P / (\sqrt{3} \times V \times \text{PF})$$

$$I = 9000 / (1.732 \times 415 \times 0.85)$$

$$I = 14.74 \text{ A}$$

$$I = 14.74 \text{ A}$$

3 Resistance of a Conductor

$$\rho = 1.72 \times 10^{-8} \text{ } \Omega \cdot \text{m}$$

$$L = 100 \text{ m}$$

$$A = 2.5 \text{ mm}^2$$

$$R = ?$$

$$R = \rho \times L / A$$

$$R = (1.72 \times 10^{-8} \times 100) / (2.5 \times 10^{-6})$$

$$R = 0.688 \text{ } \Omega$$

$$R = 0.688 \text{ } \Omega$$

4 Three Phase Apparent Power

$$V_L = 415 \text{ V}$$

$$I_L = 30 \text{ A}$$

$$S = ?$$

$$S = \sqrt{3} \times V_L \times I_L$$

$$S = 1.732 \times 415 \times 30$$

$$S = 21,556 \text{ VA}$$

$$S = 21.56 \text{ kVA}$$

$$S = 21.56 \text{ kVA}$$

5 Three Phase Real Power

$$V_L = 415 \text{ V}$$

$$I_L = 30 \text{ A}$$

$$\text{PF} = 0.9$$

$$P = ?$$

$$P = \sqrt{3} \times V_L \times I_L \times \text{PF}$$

$$P = 1.732 \times 415 \times 30 \times 0.9$$

$$P = 17,263 \text{ W}$$

$$P = 17.26 \text{ kW}$$

$$P = 17.26 \text{ kW}$$

6 Inductive Reactive Power

$$P = 15 \text{ kW}$$

$$\text{PF} = 0.8 \text{ (lagging)}$$

$$Q = ?$$

$$\phi = \cos^{-1}(0.8) = 36.87^\circ$$

$$Q = P \times \tan \phi$$

$$Q = 15 \times \tan 36.87^\circ$$

$$Q = 15 \times 0.75$$

$$Q = 11.25 \text{ kVAR}$$

$$Q = 11.25 \text{ kVAR}$$

7 Capacitor Value for Power Factor Correction

$$P = 20 \text{ kW}$$

$$\text{PF}_1 = 0.7 \text{ (lagging)}$$

$$\text{PF}_2 = 0.95 \text{ (lagging)}$$

$$V_L = 415 \text{ V}, f = 50 \text{ Hz}$$

$$C = ? \text{ (per phase)}$$

$$\tan \phi_1 = 0.714, \tan \phi_2 = 0.329$$

$$Q_c = P (\tan \phi_1 - \tan \phi_2)$$

$$Q_c = 20 (0.714 - 0.329)$$

$$Q_c = 7.70 \text{ kVAR}$$

$$C = Q_c / (2\pi f V_{ph}^2)$$

$$C = 7700 / (2\pi \times 50 \times (415/\sqrt{3})^2)$$

$$C = 45.8 \text{ } \mu\text{F (per phase)}$$

$$C = 45.8 \text{ } \mu\text{F (per phase)}$$

8 Transformer Secondary Current

$$S = 50 \text{ kVA}$$

$$V_1 = 11 \text{ kV}$$

$$V_2 = 415 \text{ V}$$

$$I_2 = ?$$

$$S = \sqrt{3} \times V_2 \times I_2$$

$$I_2 = S / (\sqrt{3} \times V_2)$$

$$I_2 = 50000 / (1.732 \times 415)$$

$$I_2 = 69.56 \text{ A}$$

$$I_2 = 69.56 \text{ A}$$



Learn • Understand • Apply • Excel



HOW TO READ CONSTRUCTION DRAWINGS

A SIMPLE GUIDE FOR CIVIL ENGINEERS & STUDENTS

1. TITLE BLOCK

Always check the title block first. It gives important information about the project.

- ★ Project Name
- ★ Drawing Title
- ★ Drawing No.
- ★ Scale
- ★ Date
- ★ Designed / Drawn by

PROJECT	RESIDENTIAL BUILDING
DRAWING TITLE	GROUND FLOOR PLAN
DRAWING NO.	A-01
SCALE	1:100
DATE	25-05-2024
DESIGNED BY	ENGR. ABC
DRAWN BY	ENGR. XYZ
CHECKED BY	ENGR. DEF

2. SCALE

Scale shows the ratio between drawing and actual size.

Common Scales

1:50	1 cm = 0.5 m
1:100	1 cm = 1 m
1:150	1 cm = 1.5 m
1:200	1 cm = 2 m

Note:

Always use scale to find actual dimensions.

3. SYMBOLS

Learn basic symbols used in drawings.

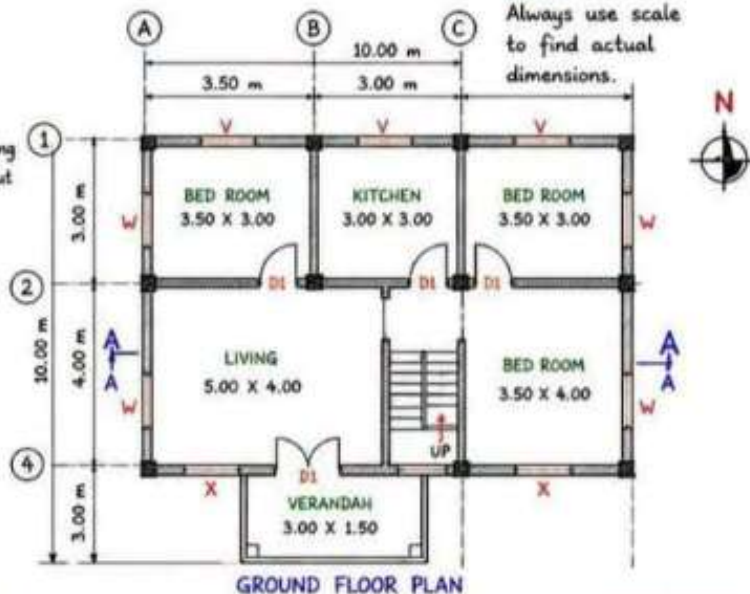
- Column
- Brick Wall
- Door
- Window
- Ventilator
- Staircase
- Center Line
- Section Mark
- Level Mark

4. PLAN (FLOOR PLAN)

Plan shows the layout of the building as if we cut it horizontally at about 1.2 m height from the floor.

It shows:

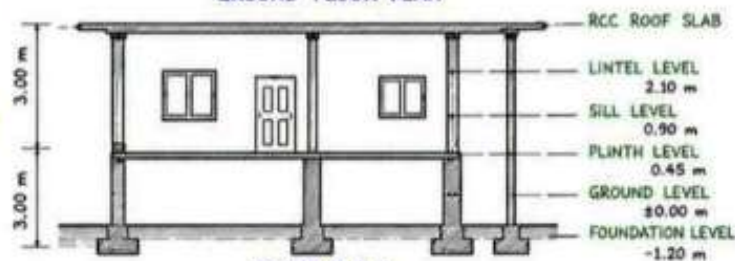
- Rooms
- Walls
- Doors
- Windows
- Columns
- Dimensions
- etc.



6. SECTION (ELEVATION CUT)

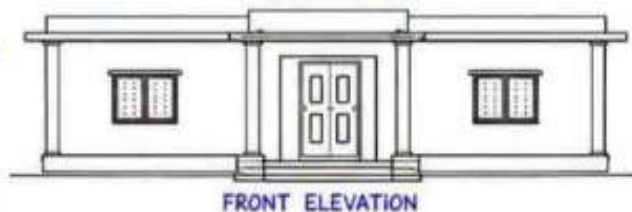
Section shows the vertical cut of the building.

It helps to understand heights and levels.



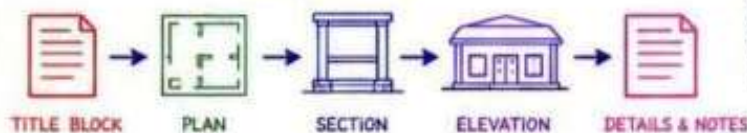
7. ELEVATION

Elevation shows the external view of the building. It helps to understand the appearance.



TIPS

- ✓ Start reading from title block.
- ✓ Understand symbols.
- ✓ Check scale and dimensions.
- ✓ Go through plan, section, elevation.
- ✓ Read notes before execution.



5. DIMENSIONS

Dimensions show the size of different elements.

- Read in meters (m)
- Always follow overall dimensions.
- Do not measure from the drawing.

Example

10.00 m = 10 meters
3.50 m = 3.5 meters

8. GENERAL NOTES

- Read all notes carefully.
- It contains important instructions.
- Follow material specifications.
- Follow workmanship guidelines.

READ DRAWINGS
CAREFULLY,
BUILD ACCURATELY."

PROSES RANGKAIAN PERKABELAN FORWARD REVERSE

1. KOMPONEN UTAMA



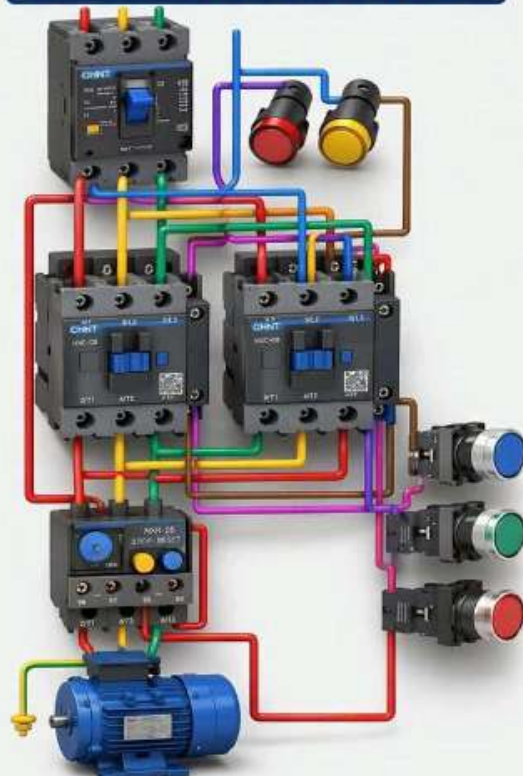
2. RANGKAIAN DAYA (POWER CIRCUIT)



3. RANGKAIAN KONTROL FORWARD



4. RANGKAIAN KONTROL REVERSE



KETERANGAN PROSES

1. Komponen Utama disiapkan sesuai fungsi masing-masing.
2. Rangkaian Daya menghubungkan sumber 3 fasa ke kontaktor, TOR, dan motor.
3. Rangkaian Kontrol Forward mengaktifkan kontaktor F untuk putaran maju.
4. Rangkaian Kontrol Reverse mengaktifkan kontaktor R untuk putaran mundur.

- | | |
|---------------------------|---------------------|
| R (L1) - Merah | Netral - Biru |
| S (L2) - Kuning | Control tambahan |
| T (L3) - Hijau | Lainnya (Interlock) |
| Kontrol - Merah / Lainnya | Ground (PE) |

ELECTRICAL CALCULATION



By Electrical Knowledge

1 Ohm's Law

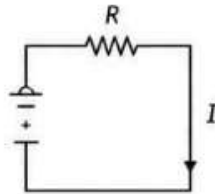
$$V = I \times R$$

V = Voltage (V)
I = Current (A)
R = Resistance (Ω)

Example:

If $I = 5 \text{ A}$, $R = 12 \text{ } \Omega$

$$V = 5 \times 12 = \boxed{60 \text{ V}}$$



2 Power

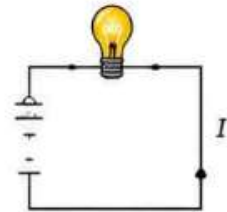
$$P = V \times I$$

P = Power (W)
V = Voltage (V)
I = Current (A)

Example:

If $V = 230 \text{ V}$, $I = 4 \text{ A}$

$$P = 230 \times 4 = \boxed{920 \text{ W}}$$



3 Resistance

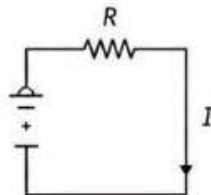
$$R = \frac{V}{I}$$

R = Resistance (Ω)
V = Voltage (V)
I = Current (A)

Example:

If $V = 48 \text{ V}$, $I = 6 \text{ A}$

$$R = 48 / 6 = \boxed{8 \text{ } \Omega}$$



4 Energy

$$E = P \times t$$

E = Energy (J or kWh)
P = Power (W or kW)
t = Time (s or h)

Example:

If $P = 2 \text{ kW}$, $t = 3 \text{ h}$

$$E = 2 \times 3 = \boxed{6 \text{ kWh}}$$



5 Series Resistance

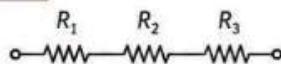
$$R_s = R_1 + R_2 + R_3 + \dots$$

R_s = Total Resistance (Ω)
 R_1, R_2, R_3 = Individual Resistances (Ω)

Example:

If $R_1 = 2 \text{ } \Omega$, $R_2 = 3 \text{ } \Omega$, $R_3 = 5 \text{ } \Omega$

$$R_s = 2 + 3 + 5 = \boxed{10 \text{ } \Omega}$$



6 Parallel Resistance

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

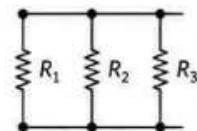
R_p = Total Resistance (Ω)
 R_1, R_2, R_3 = Individual Resistances (Ω)

Example:

If $R_1 = 6 \text{ } \Omega$, $R_2 = 3 \text{ } \Omega$, $R_3 = 2 \text{ } \Omega$

$$\frac{1}{R_p} = \frac{1}{6} + \frac{1}{3} + \frac{1}{2} = \frac{1+2+3}{6} = \frac{6}{6} = 1$$

$$R_p = \boxed{1 \text{ } \Omega}$$



7 Power Factor

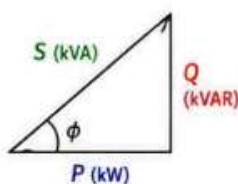
$$PF = \frac{P}{S} = \cos \phi$$

PF = Power Factor (Unitless)
P = Real Power (kW)
S = Apparent Power (kVA)
 ϕ = Phase Angle (Degree)

Example:

If $P = 8 \text{ kW}$, $S = 10 \text{ kVA}$

$$PF = \frac{8}{10} = \boxed{0.8 \text{ (lagging)}}$$



8 Voltage Drop

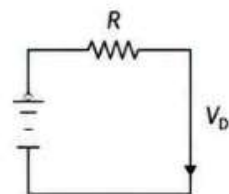
$$V_D = I \times R$$

V_D = Voltage Drop (V)
I = Current (A)
R = Resistance (Ω)

Example:

If $I = 10 \text{ A}$, $R = 2 \text{ } \Omega$

$$V_D = 10 \times 2 = \boxed{20 \text{ V}}$$

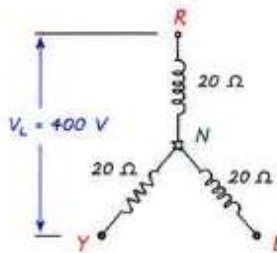


Learn • Understand • Apply • Excel



STAR (Y) AND DELTA (Δ) CONNECTIONS - QUESTIONS & ANSWERS

- Q1** A 3-phase balanced load of 3 resistors of $20\ \Omega$ each is connected in star (Y) to a 3-phase, 400 V, 50 Hz supply. Find (i) Line current (I_L) (ii) Phase current (I_{ph}) (iii) Power consumed.

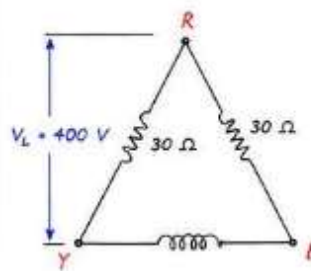


Solution :

- (i) In star connection, $V_{ph} = \frac{V_L}{\sqrt{3}}$
 $= \frac{400}{\sqrt{3}} = 231\text{ V}$
 $I_{ph} = \frac{V_{ph}}{R_{ph}} = \frac{231}{20} = 11.55\text{ A}$
 $I_L = I_{ph} = 11.55\text{ A}$
 (ii) $I_{ph} = 11.55\text{ A}$; $I_L = 11.55\text{ A}$
 (iii) Power, $P = \sqrt{3} V_L I_L \cos \phi$
 Assume $\cos \phi = 1$ (pure resistive)
 $P = \sqrt{3} \times 400 \times 11.55 \times 1$
 $= 8000\text{ W} = 8\text{ kW}$

Answer : (i) $I_L = 11.55\text{ A}$ (ii) $I_{ph} = 11.55\text{ A}$ (iii) $P = 8\text{ kW}$

- Q2** A 3-phase balanced load of 3 resistors of $30\ \Omega$ each is connected in delta (Δ) to a 3-phase, 400 V, 50 Hz supply. Find (i) Line current (I_L) (ii) Phase current (I_{ph}) (iii) Power consumed.

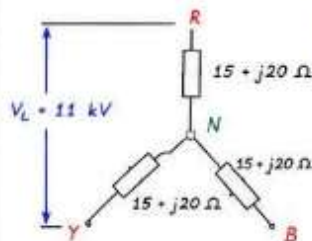


Solution :

- In delta connection, $V_{ph} = V_L = 400\text{ V}$
 $I_{ph} = \frac{V_{ph}}{R_{ph}} = \frac{400}{30} = 13.33\text{ A}$
 $I_L = \sqrt{3} I_{ph} = \sqrt{3} \times 13.33$
 $= 23.09\text{ A}$
 Power, $P = \sqrt{3} V_L I_L \cos \phi$
 $= \sqrt{3} \times 400 \times 23.09 \times 1$
 $= 16000\text{ W} = 16\text{ kW}$

Answer : (i) $I_L = 23.09\text{ A}$ (ii) $I_{ph} = 13.33\text{ A}$ (iii) $P = 16\text{ kW}$

- Q3** A 3-phase, 11 kV (line to line), 50 Hz supply is given to a balanced star (Y) connected load. Each phase impedance is $15 + j20\ \Omega$. Find (i) Phase voltage (V_{ph}) (ii) Phase current (I_{ph}) (iii) Line current (I_L) (iv) Power factor.



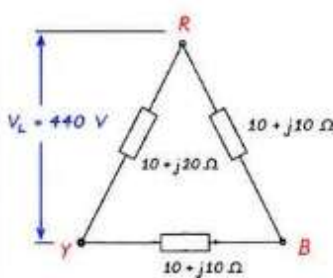
Solution :

- (i) In star, $V_{ph} = \frac{V_L}{\sqrt{3}}$
 $= \frac{11000}{\sqrt{3}} = 6350\text{ V}$
 (ii) $|Z_{ph}| = \sqrt{15^2 + 20^2} = 25\ \Omega$
 $\phi = \tan^{-1} \left(\frac{20}{15} \right) = 53.13^\circ$
 $I_{ph} = \frac{V_{ph}}{|Z_{ph}|} = \frac{6350}{25} = 254\text{ A}$
 (iii) $I_L = I_{ph} = 254\text{ A}$

(iv) Power factor, $\cos \phi = \cos 53.13^\circ = 0.6$ (lagging)

Answer : (i) $V_{ph} = 6350\text{ V}$ (ii) $I_{ph} = 254\text{ A}$ (iii) $I_L = 254\text{ A}$
 (iv) pf = 0.6 lagging

- Q4** A 3-phase, 440 V (line to line), 50 Hz supply is given to a balanced delta (Δ) connected load. Each phase impedance is $10 + j10\ \Omega$. Find (i) Phase current (I_{ph}) (ii) Line current (I_L) (iii) Power factor.



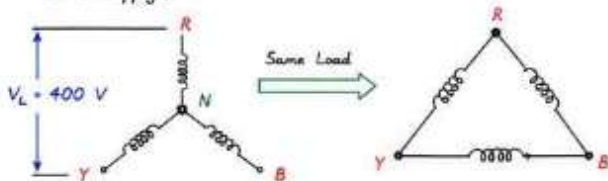
Solution :

- In delta, $V_{ph} = V_L = 440\text{ V}$
 $|Z_{ph}| = \sqrt{10^2 + 10^2} = 14.14\ \Omega$
 $\phi = \tan^{-1} \left(\frac{10}{10} \right) = 45^\circ$
 (i) $I_{ph} = \frac{V_{ph}}{|Z_{ph}|} = \frac{440}{14.14} = 31.11\text{ A}$
 (ii) $I_L = \sqrt{3} I_{ph}$
 $= \sqrt{3} \times 31.11 = 53.87\text{ A}$

(iii) Power factor, $\cos \phi = \cos 45^\circ = 0.707$ lagging

Answer : (i) $I_{ph} = 31.11\text{ A}$ (ii) $I_L = 53.87\text{ A}$ (iii) pf = 0.707 lagging

- Q5** A balanced 3-phase load takes 10 A per line when connected in star to a 400 V, 3-phase supply. What will be the line current when the same load is connected in delta to the same supply?



Solution :

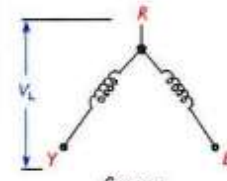
In star connection, $I_{L(star)} = 10\text{ A}$

For same load, in delta connection :

$$I_{L(delta)} = \sqrt{3} \times I_{L(star)} = \sqrt{3} \times 10 = 17.32\text{ A}$$

Answer : Line current in delta connection = 17.32 A

- Q6** A 3-phase motor is designed to operate on 400 V line voltage. During starting it is connected in star and after starting it is connected in delta. How many times is the starting current reduced in star connection compared to delta connection?



Solution :

- In star, $V_{ph} = \frac{V_L}{\sqrt{3}}$, In delta, $V_{ph} = V_L$
 Since current $\propto$ voltage (for same impedance)
 Starting current in star, $I_{star} \propto \frac{V_L}{\sqrt{3}}$
 Starting current in delta, $I_{delta} \propto V_L$

Compare Current

$$\frac{I_{star}}{I_{delta}} = \frac{V_L/\sqrt{3}}{V_L} = \frac{1}{\sqrt{3}}$$

$\therefore$ Current in star is $\frac{1}{\sqrt{3}}$ times of delta
 or delta current is $\sqrt{3}$ times of star current.

Answer : Starting current in star is $\frac{1}{\sqrt{3}}$ times of delta current.

IMPORTANT FORMULAS

For Star (Y) Connection

$$V_L = \sqrt{3} V_{ph}$$

$$I_L = I_{ph}$$

$$P = \sqrt{3} V_L I_L \cos \phi$$

For Delta (Δ) Connection

$$V_L = V_{ph}$$

$$I_L = \sqrt{3} I_{ph}$$

$$P = \sqrt{3} V_L I_L \cos \phi$$

KEY POINTS

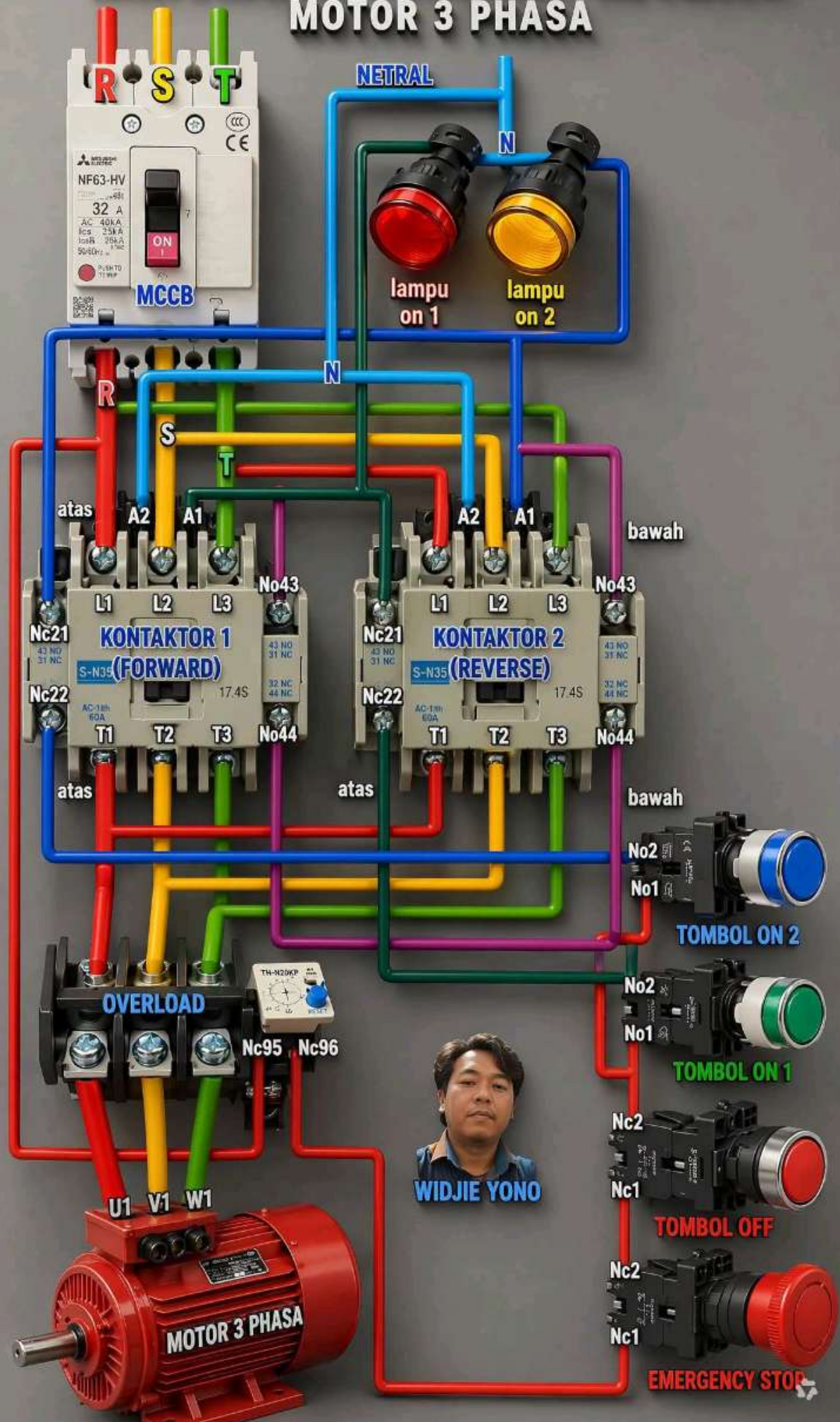
- In star connection, neutral point is available.
- In delta connection, there is no neutral point.
- For same load and same line voltage, $I_{L(delta)} = \sqrt{3} I_{L(star)}$
- For same power, line current in star is less than delta.
- Star connection is used for high voltage side.
- Delta connection is used for low voltage side.

REMEMBER

Δ for high current (because I_L larger)
 Y for high voltage (because V_{ph} less)
 Star gives economy in winding insulation.
 Delta gives more power for same kVA.



RANGKAIAN FORWARD REVERSE MOTOR 3 PHASA





Secure & Transparent

Advanced security protocols



Energy-Efficient Infrastructure

Sustainable by design



Global Network Reliability

High-performance network with 99.9% uptime



**GROW WITH TECHNOLOGY.
EARN WITH CONFIDENCE.**

Join a Global Community
Building the Future
Together.



Advanced
Security Protocols



Distributed
Network
Technology



Sustainable &
Eco-Friendly

MODEL: A2



Hashrate	226 TH/s
Power	3500 W
Energy Efficiency	15.5 J/TH
Noise Level	≤ 75 dB
Cooling System	Advanced Air Cooling



Real-time
Monitoring



Automatic
Reward
Distribution



Referral Rewards
Program



User-Friendly
Experience



Referral Rewards for Mining Pool

Upgrade Your Referral
Privileges
Enjoy Up to 20% Referral
Rewards

[View Details >](#)



Notification



Online Service



About



Invite

A2

+7.00%

Mining In Process



1



Claude Pro

Harga: Mulai Rp125.000

Durasi: 1 Bulan / 3 Bulan

Garansi: Full garansi

2



Claude Max

Harga: Mulai Rp175.000

Durasi: 1 Bulan / 3 Bulan / 6 Bulan

Garansi: Full garansi

3



Gemini Pro

Harga: Mulai Rp120.000

Durasi: 18 Bulan / 48 Bulan

Garansi: Full garansi

4



Gemini Ultra

Harga: Mulai Rp115.000

Durasi: 1 Bulan

Garansi: Full garansi

5



ChatGPT Plus

Harga: Mulai Rp75.000

Durasi: 1 Bulan / 3 Bulan

Garansi: Full garansi

6



ChatGPT Pro

Harga: Mulai Rp135.000

Durasi: 1 Bulan / 3 Bulan

Garansi: Full garansi

7



Kiro Pro

Harga: Mulai Rp60.000

Durasi: 1 Bulan

Garansi: Full garansi

8



Kiro Power

Harga: Mulai Rp100.000

Durasi: 1 Bulan

Garansi: Full garansi

9



Leonardo.Ai

Harga: Mulai Rp75.000

Durasi: 1 Bulan / 3 Bulan

Garansi: Full garansi

10



A2E AI

Harga: Mulai Rp100.000

Durasi: 1 Bulan

Garansi: Full garansi

11



SuperGrok

Harga: Mulai Rp80.000

Durasi: 1 Bulan / 3 Bulan

Garansi: Full garansi

12



Seedance 2.0

Harga: Mulai Rp100.000

Durasi: 1 Bulan

Garansi: Full garansi

13



Dola unlimited

Harga: Mulai Rp100.000

Durasi: 1 Bulan

Garansi: Full garansi

14



Cursor

Harga: Mulai Rp125.000

Durasi: 1 Bulan

Garansi: Full garansi

15



Kling AI

Harga: Mulai Rp155.000

Durasi: 1 Bulan

Garansi: Full garansi

16



Xiaomi MiMo Plan Max

Harga: Mulai Rp150.000

Durasi: 1 Bulan

Garansi: Full garansi

17



DeepSeek

Harga: Mulai Rp130.000

Durasi: 1 Bulan

Garansi: Full garansi

18



GitHub Copilot

Harga: Mulai Rp135.000

Durasi: 1 Bulan / 3 Bulan

Garansi: Full garansi



⚡ Infrastruktur untuk Pusat Data ... lainnya



Smart Infrastructure for Data Centers

Integrated systems for reliable deployment.



[Learn More →](#)





DUCT SIZING SELECTION – HVAC



Right Size – Right Air Flow – Right Performance

1. DESIGN DATA (INPUT)

- Air Flow (Q) : CFM / m³/h
- Duct Type : Rectangular / Round
- Available Space : Width (W) x Height (H)
- Friction Loss : As per design
- Maximum Velocity : As per duct location
- Static Pressure : As per system

2. BASIC FORMULAS

Air Flow Formula

$$Q = A \times V$$

Where,

Q = Air Flow (CFM or m³/h)

A = Cross sectional area (ft² or m²)

V = Velocity in duct (fpm or m/s)

Rectangular Duct

$$A = W \times H$$

W = Width (ft or in)

H = Height (ft or in)

Round Duct

$$A = \frac{\pi D^2}{4}$$

D = Diameter (ft or in)

$\pi = 3.1416$

3. RECOMMENDED VELOCITY RANGE

Duct Type / Location	Velocity	
	m/s	fpm
Main Supply Duct	6 – 9	1200 – 1800
Branch Supply Duct	4 – 6	800 – 1200
Main Return Duct	5 – 8	1000 – 1600
Branch Return Duct	3 – 5	600 – 1000
Exhaust Duct	6 – 10	1200 – 2000
Fresh Air Duct	5 – 8	1000 – 1600

GOOD PRACTICE

- Use Lower velocity for quiet operation
- Use Higher velocity for smaller duct size
- Keep velocity within recommended range
- Balance between size, noise and pressure drop

4. DUCT SIZE CALCULATION STEPS

- Select Velocity (V)
Choose velocity as per recommended range.
- Calculate Required Area (A)
$$A = \frac{Q}{V}$$
- Calculate Duct Dimension
 - Rectangular Duct : $A = W \times H$
Choose W/H ratio (1:1 to 4:1 preferred)
 - Round Duct : $A = \frac{\pi D^2}{4}$
- Select Standard Size
From standard duct size table.
(Select next higher size than calculated)

5. EXAMPLE – RECTANGULAR DUCT

Given:

- Air Flow (Q) = 2500 CFM
- Selected Velocity (V) = 1000 fpm
- W/H Ratio = 2 : 1

Step 1: Calculate Area

$$A = \frac{Q}{V} = \frac{2500}{1000} = 2.50 \text{ ft}^2$$

Step 2: Assume W = 2H (Ratio 2:1)

Let H = x ft, W = 2x ft

$$A = W \times H = 2x \times x = 2x^2 = 2.50$$

$$x^2 = 1.25 \Rightarrow x = 1.118 \text{ ft}$$

H = 1.118 ft (13.4 in)

W = 2x = 2.236 ft (26.8 in)

Selected Duct Size (Standard)

28" (W) x 14" (H)



6. EXAMPLE – ROUND DUCT

Given:

- Air Flow (Q) = 2600 CFM
- Selected Velocity (V) = 1200 fpm

Step 1: Calculate Area

$$A = \frac{Q}{V} = \frac{2600}{1200} = 2.167 \text{ ft}^2$$

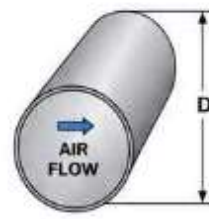
Step 2: Calculate Diameter

$$D = \sqrt{\frac{4A}{\pi}} = \sqrt{\frac{4 \times 2.167}{3.1416}}$$

$$D = \sqrt{2.760} = 1.661 \text{ ft} = 19.9 \text{ in}$$

Selected Duct Size (Standard)

20" Ø



7. RECTANGULAR DUCT SIZE TABLE (CFM) – VELOCITY IN FPM

Size W" x H"	Velocity (FPM)					
	600	800	1000	1200	1400	1600
8" x 6"	215	286	358	429	501	572
10" x 6"	269	359	448	538	628	717
12" x 8"	430	573	717	860	1004	1147
14" x 10"	597	796	995	1194	1393	1592
16" x 10"	682	909	1136	1364	1591	1818
18" x 12"	860	1147	1434	1720	2007	2294
20" x 12"	956	1275	1594	1913	2232	2549
22" x 14"	1176	1568	1960	2352	2744	3136
24" x 16"	1529	2038	2548	3058	3567	4077

Note: Values are approximate.

8. ROUND DUCT SIZE TABLE (CFM) – VELOCITY IN FPM

Diameter (inch)	Velocity (FPM)					
	600	800	1000	1200	1400	1600
10	235	314	392	471	549	628
12	338	452	565	678	791	904
14	461	615	768	922	1075	1229
16	602	803	1003	1204	1405	1606
18	760	1014	1267	1521	1775	2028
20	941	1255	1569	1883	2198	2510
22	1135	1513	1892	2270	2649	3027
24	1352	1803	2253	2704	3155	3606
26	1584	2112	2640	3168	3696	4224

Note: Values are approximate.

9. VELOCITY TO NOISE REFERENCE

Velocity (FPM)	Noise Level	Application / Area
Below 700	Very Low	Recording Studio, Library, Hospital
700 – 1000	Low	Offices, Residential
1000 – 1500	Medium	Shops, Commercial Building
1500 – 2000	High	Industrial Areas
Above 2000	Very High	Avoid (Excess Noise)

10. PRESSURE DROP CHECK (DUCT FRICTION LOSS)

Friction Loss Formula

$$\Delta P = f \times \frac{L}{D_e} \times \frac{\rho \times V^2}{2}$$

Where,

- ΔP = Friction loss (Pa)
- f = Friction factor (from chart)
- L = Duct length (m)
- D_e = Equivalent diameter (m)
- ρ = Air density (1.2 kg/m³ approx.)
- V = Velocity (m/s)

Good Practice

- Keep friction loss in mains: 0.5 – 1.2 Pa/m
- Keep friction loss in branches: 0.3 – 0.6 Pa/m
- Total external static pressure = Sum of all losses

13. GENERAL TIPS FOR DUCT SIZING

- Fix velocity first, then calculate size.
- Select standard size (next higher).
- Maintain proper W/H ratio (1:1 to 4:1).
- Avoid too many bends and obstructions.
- Provide balancing dampers in branches.
- Insulate ducts as per requirement.
- Check pressure drop to ensure fan performance.

11. EQUIVALENT DIAMETER (RECTANGULAR DUCT)

Use for pressure drop calculation

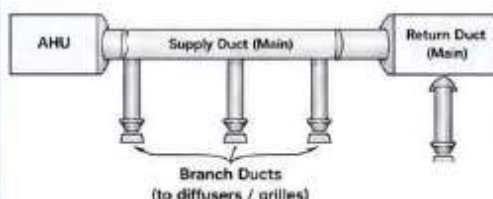
$$D_e = \frac{1.30 (W \times H)^{0.625}}{(W + H)^{0.25}}$$

Where,

W = Width (m)

H = Height (m)

12. TYPICAL DUCT LAYOUT



14. DUCT SHAPE RATIO GUIDE (RECTANGULAR)

W/H Ratio	1 : 1	2 : 1	3 : 1	4 : 1
Air Flow Capacity	Good	Better	Good	Acceptable

⚠ Prefer W/H ratio between 1:1 to 4:1 for better performance.

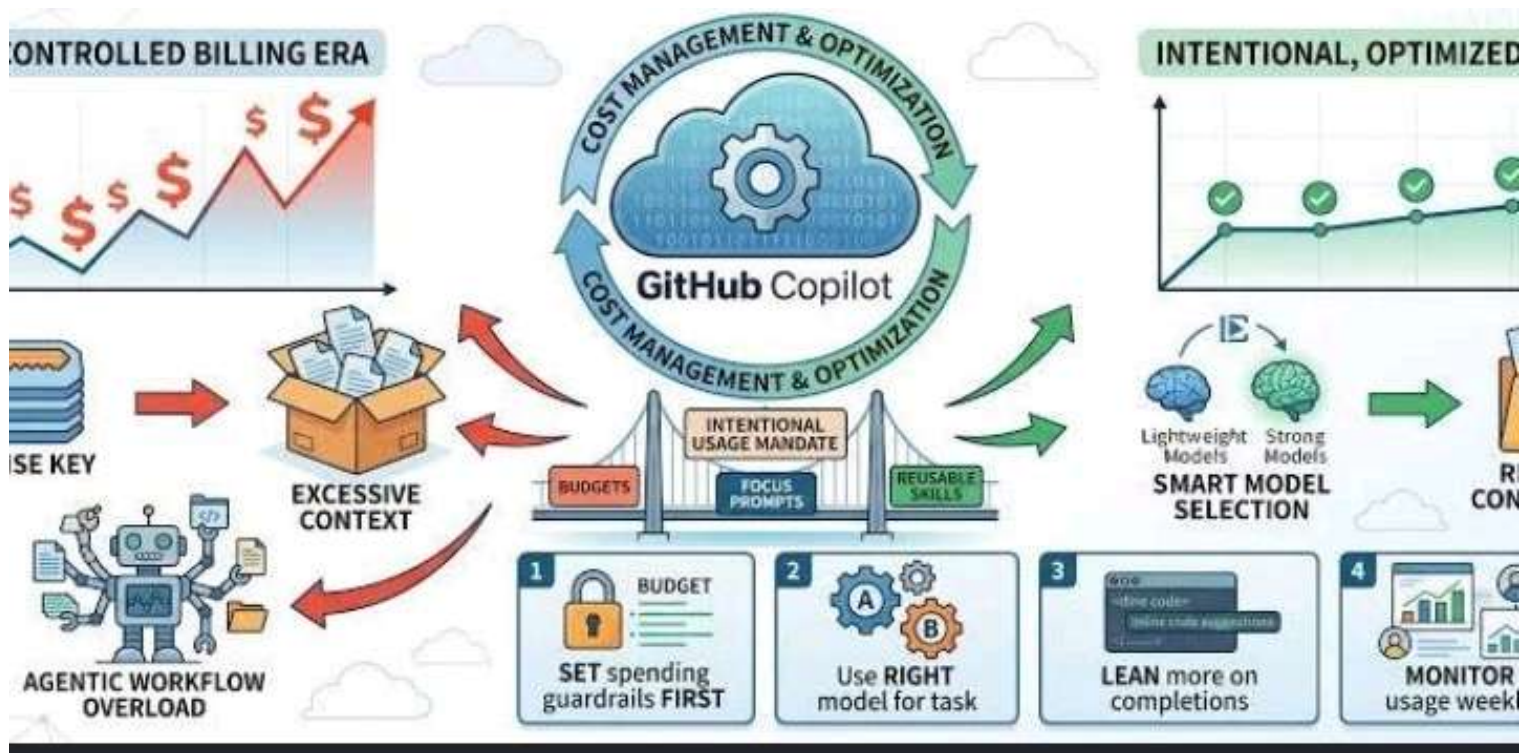
KEY

- Q = Air Flow (CFM)
- A = Area (ft²)
- V = Velocity (fpm)
- W = Width (inch)
- H = Height (inch)
- D = Diameter (inch)
- ΔP = Pressure Drop (Pa)

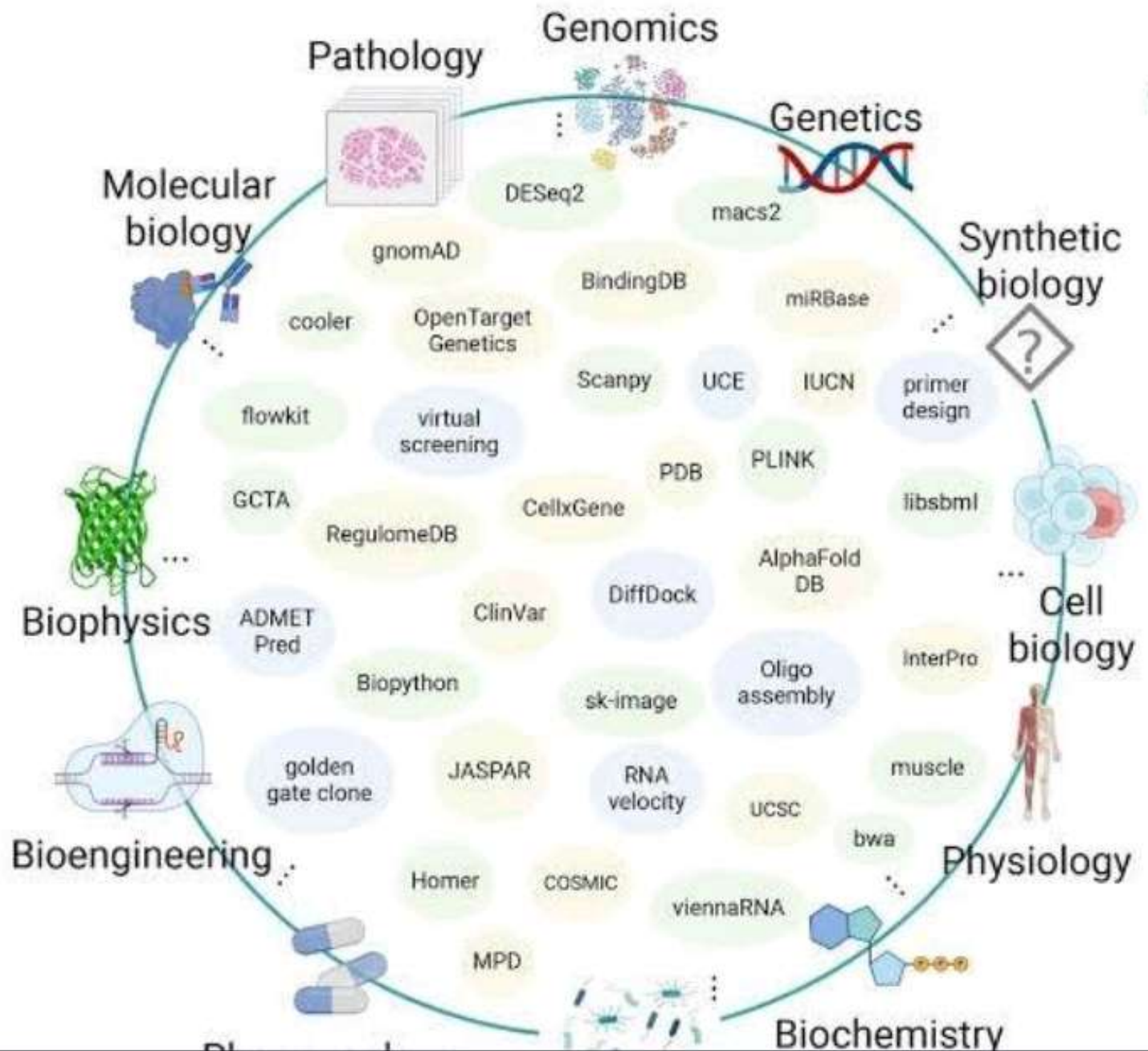
IMPORTANT NOTE

- Proper duct sizing ensures required airflow, low noise, low power consumption and efficient HVAC performance.
- Final sizing must be verified with pressure drop calculations and site conditions.

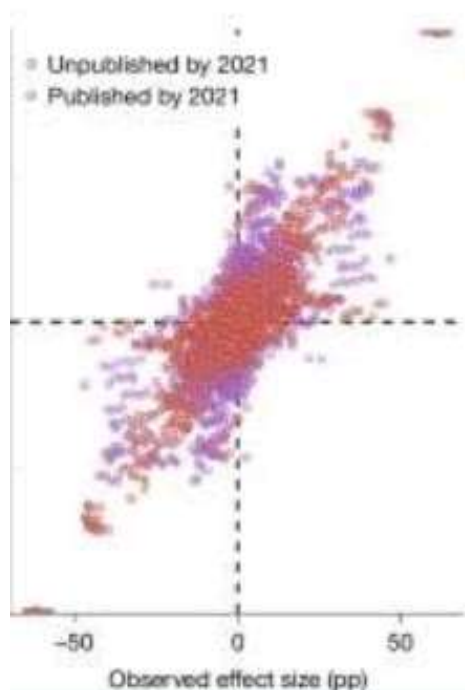




An AI Agent Environment with Unified Biomedical Action Space

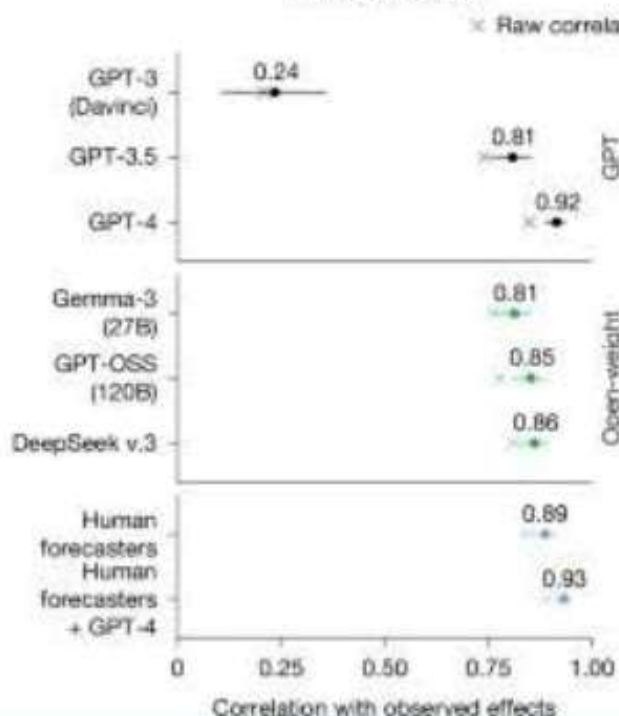


All effects (GPT-4)



D

Model comparison



C



